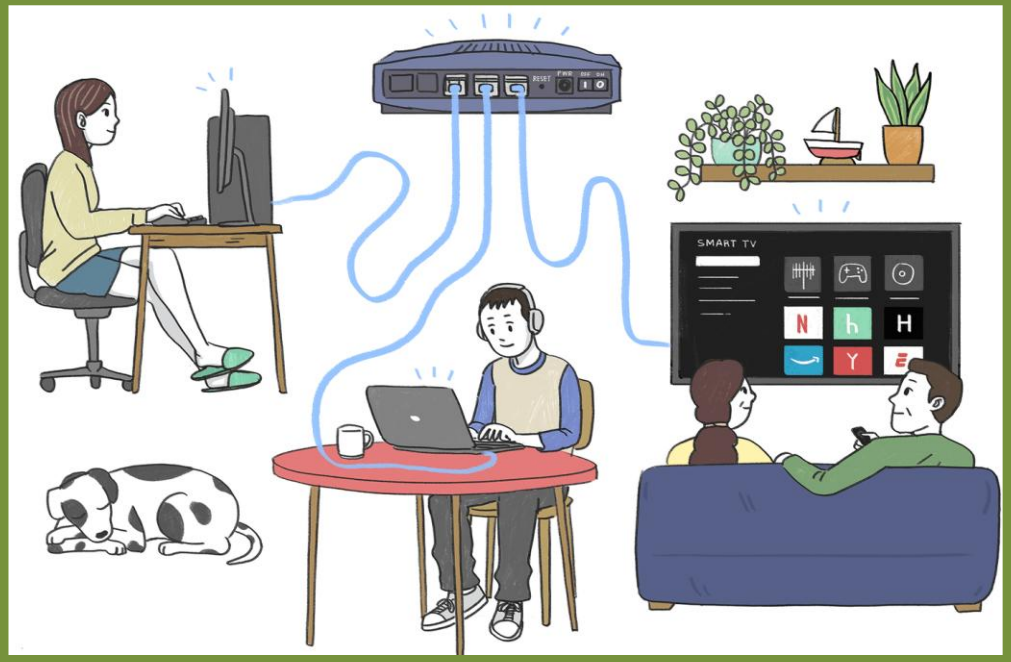
Local Area Network: Ethernet



Contents

- Introduction - IEEE
- 802.3 IEEE standard
- Token Ring: 802.5 IEEE standard
- Token Bus: 802.4 IEEE standard
- Wireless LAN Protocols

IEEE

- Institution of Electrical and Electronic Engineering (IEEE)
- A professional non-profit organization
- Project group 802 under IEEE
  - *Entrusted with the task of setting standards relating to physical and logical links of nodes in a network*
- Standard mostly applies to the Physical and Data Link layers
- Example
  - *IEEE 802.3 standard for the Ethernet bus network*

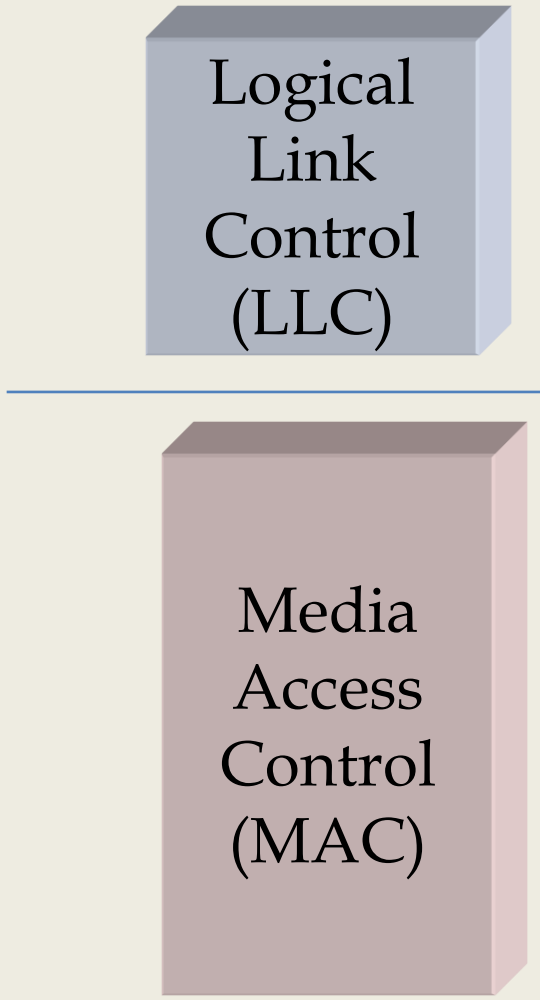
IEEE 802 Focus

- OSI Reference
  - *Data Link layer*
  - *Physical layer*
- Areas of applications
  - *Network cards and cables*
  - *WAN connectivity etc.*
- Different subgroups under 802 that focus on different activities of the LAN

- IEEE 802.1 Internetworking
- IEEE 802.2 Logical Link Control
- IEEE 802.3 CSMA/CD
- IEEE 802.4 Token Bus LAN
- IEEE 802.5 Token Ring LAN
- IEEE 802.6 Metropolitan Area Network

- IEEE 802.7 Broadband Technical Advisory Group
- IEEE 802.8 Fiber optic Technical Advisory Group
- IEEE 802.9 Integrated Voice/Data Networks
- IEEE 802.10 Network Security
- IEEE 802.11 Wireless Networks
- IEEE 802.12 Demand Priority Access LANs

Perspective of IEEE 802 Standards in Network Communication

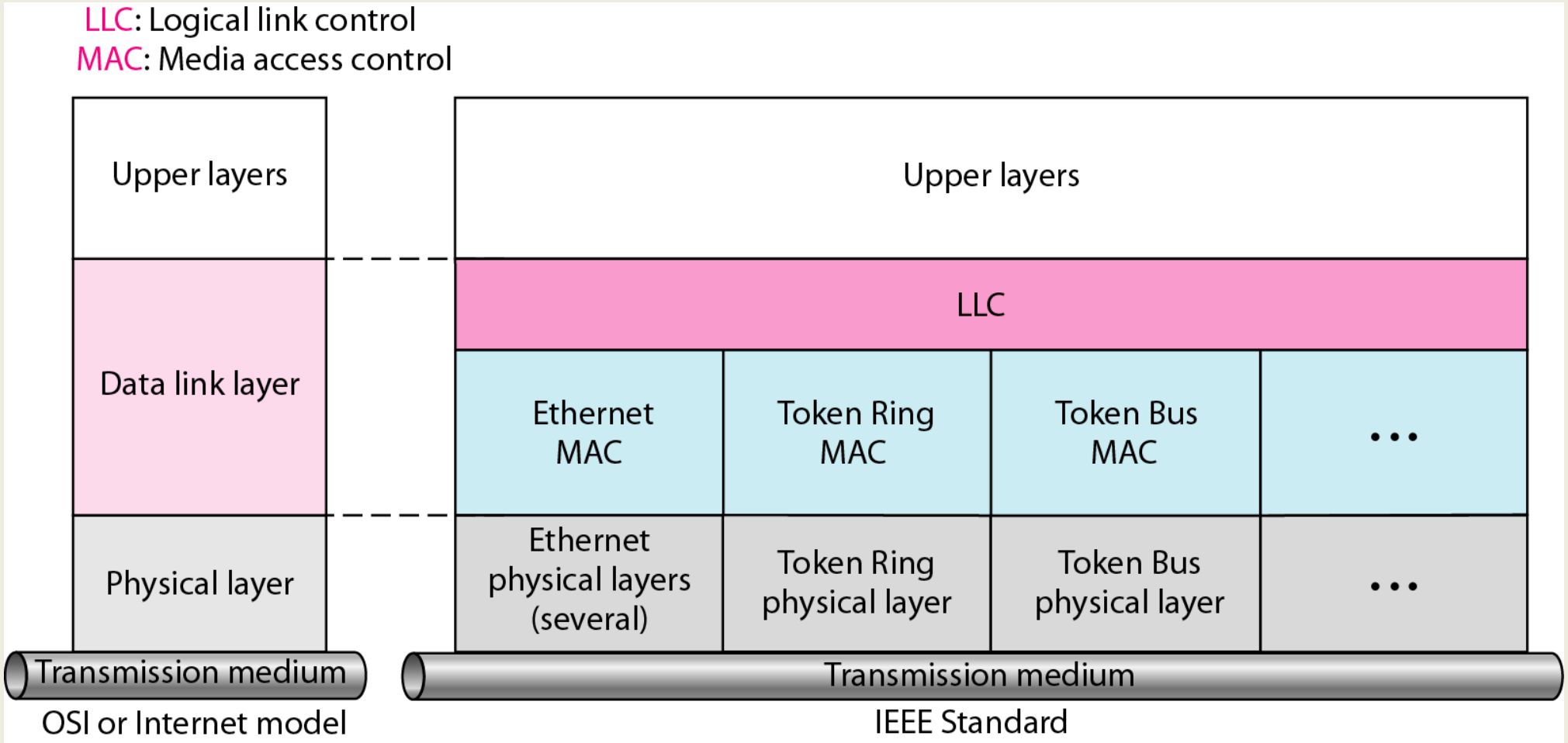


802.2

802.3
802.4
802.5
802.12

802.1 applies to both.

IEEE standard for LANs

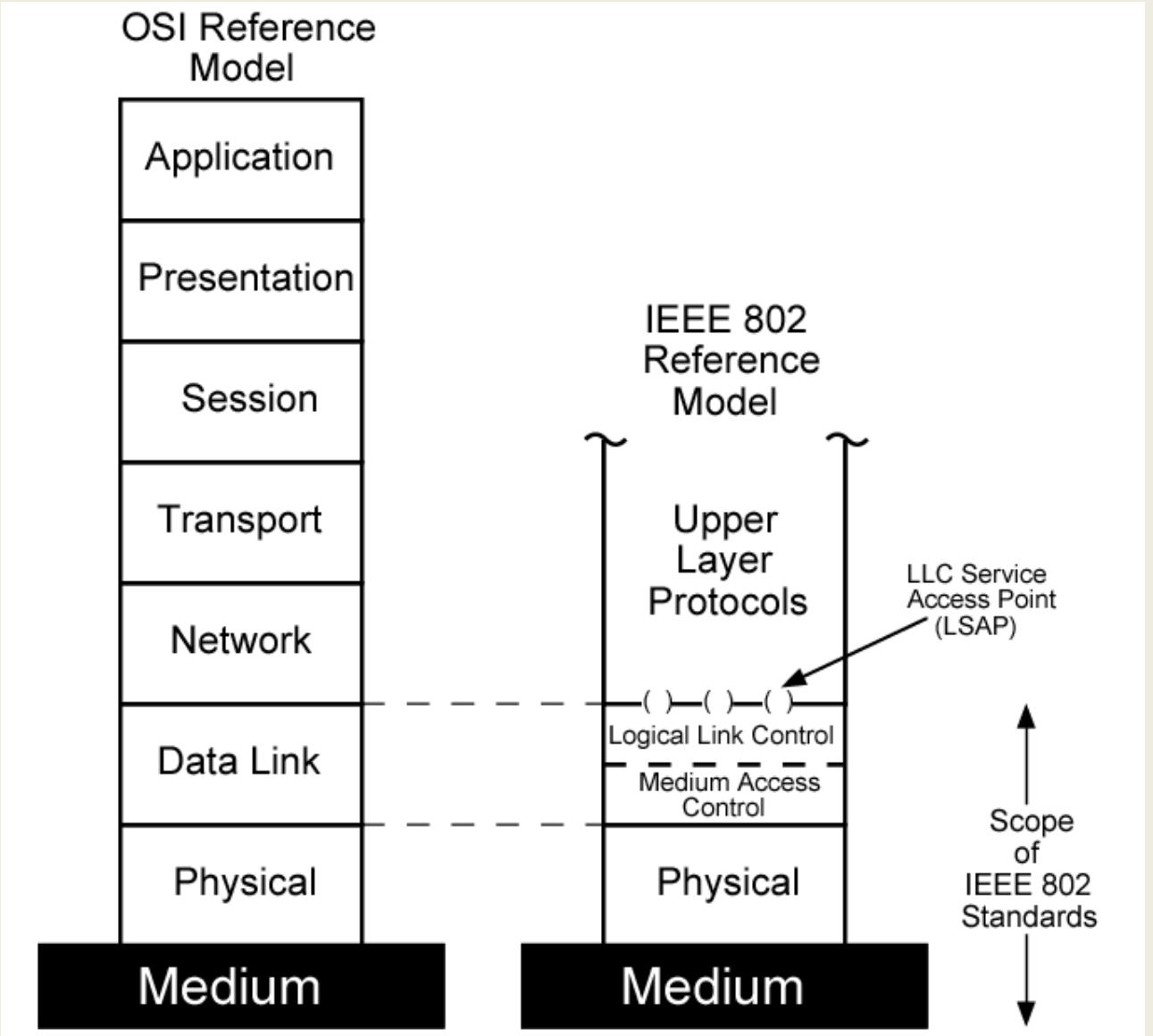


Logical Link Control (LLC)

- In IEEE Project 802, flow control, error control, and part of the framing duties are collected into one sublayer called the **logical link control (LLC)**.
- Framing is handled in both the LLC sublayer and the MAC sublayer.
- The LLC provides a single link-layer control protocol for all IEEE LANs.
- LLC protocol can provide interconnectivity between different LANs because it makes the MAC sublayer transparent.

Media Access Control (MAC)

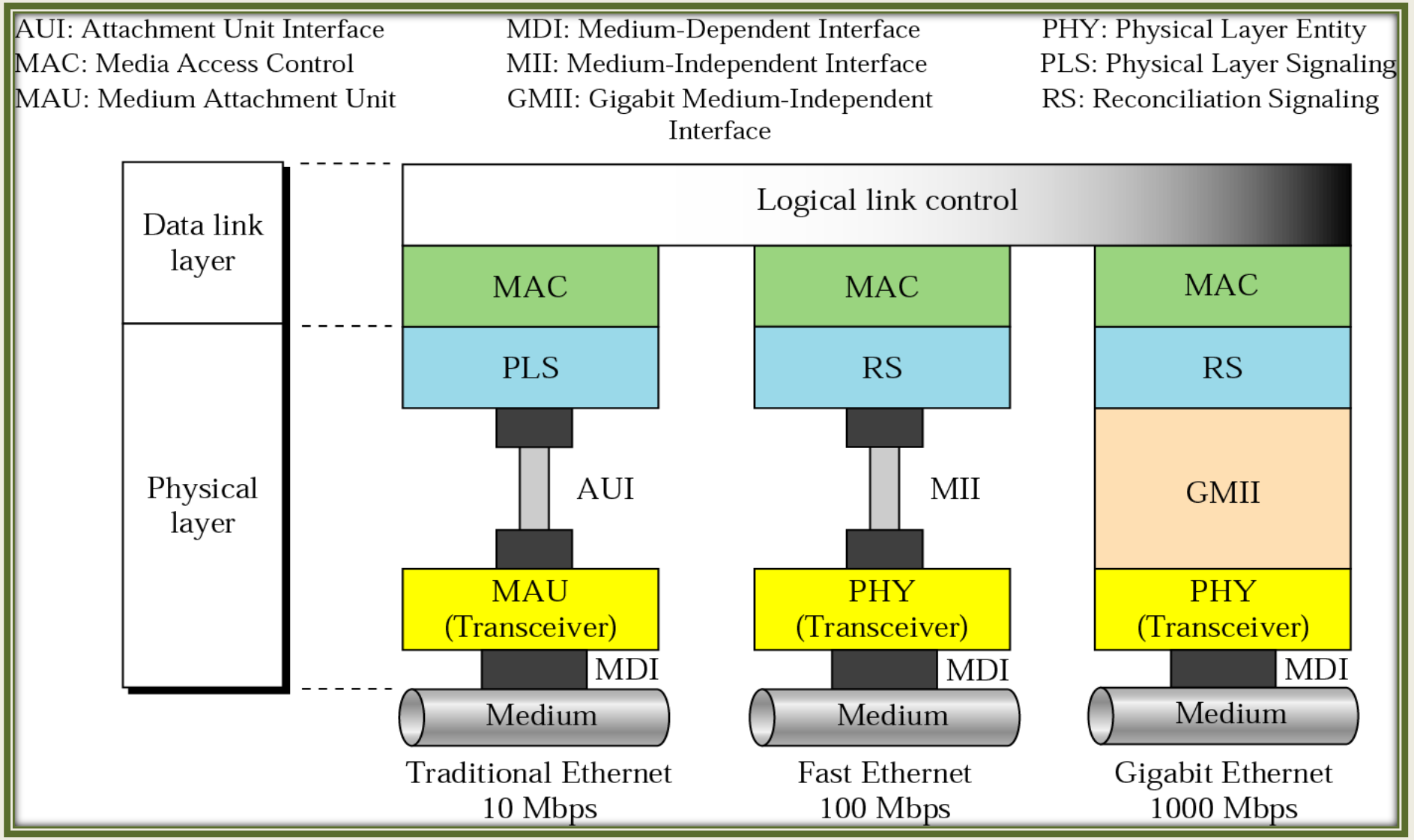
- IEEE Project 802 has created a sublayer called media access control that defines the specific access method for each LAN.
- For example, it defines CSMA/CD as the media access method for Ethernet LANs and defines the token-passing method for Token Ring and Token Bus LANs.

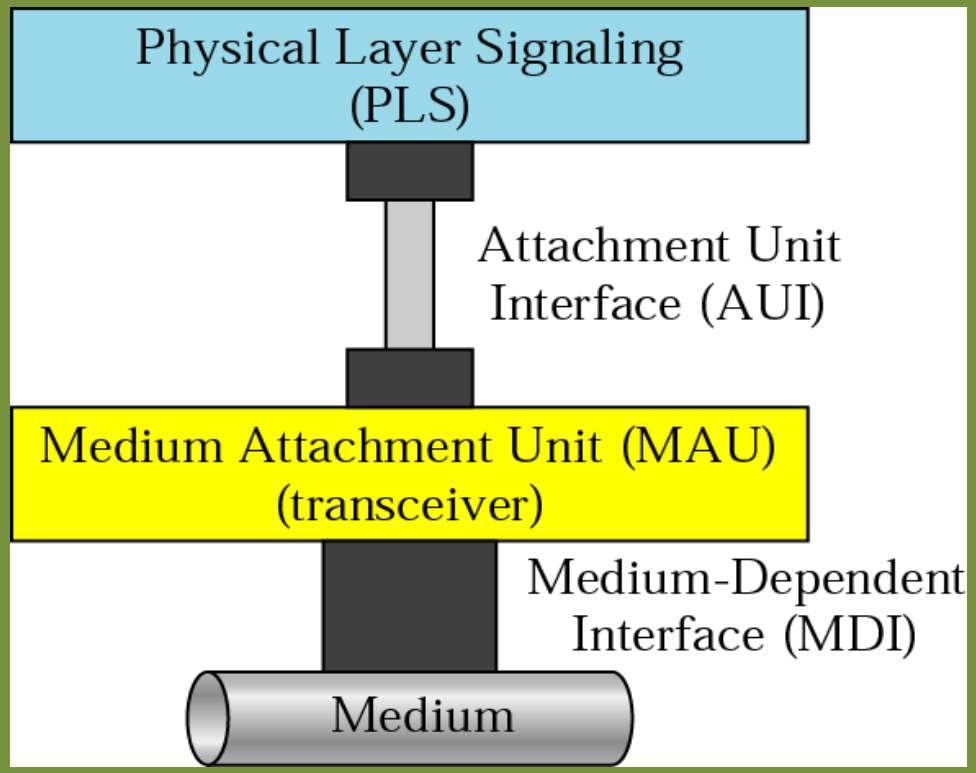

OSI Reference Model
Application
Presentation
Session
Transport
Network
Data Link
Physical
Medium
IEEE 802 Reference Model
Upper Layer Protocols
LLC Service Access Point (LSAP)
Logical Link Control
Medium Access Control
Physical
Medium
Scope of IEEE 802 Standards

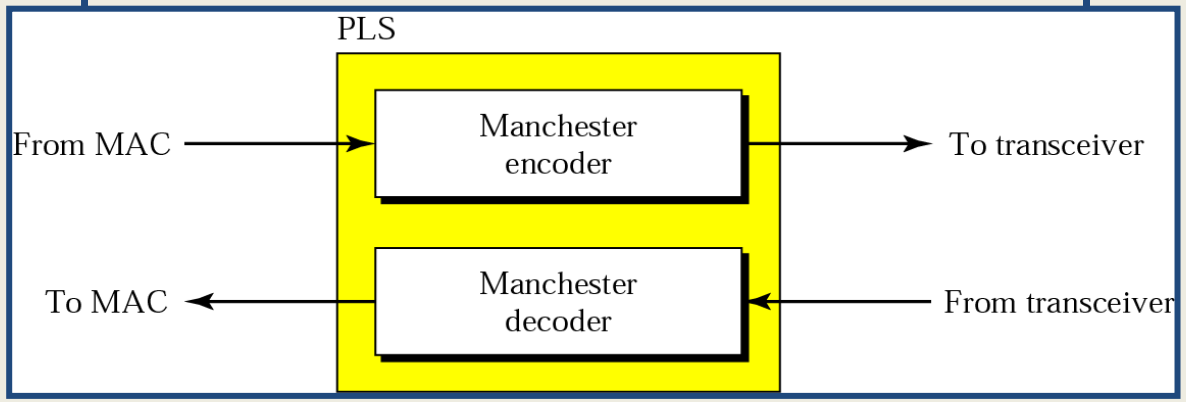
IEEE 802 Layers - Physical

- Encoding/decoding
- Preamble generation/removal
- Bit transmission/reception
- Transmission medium and topology

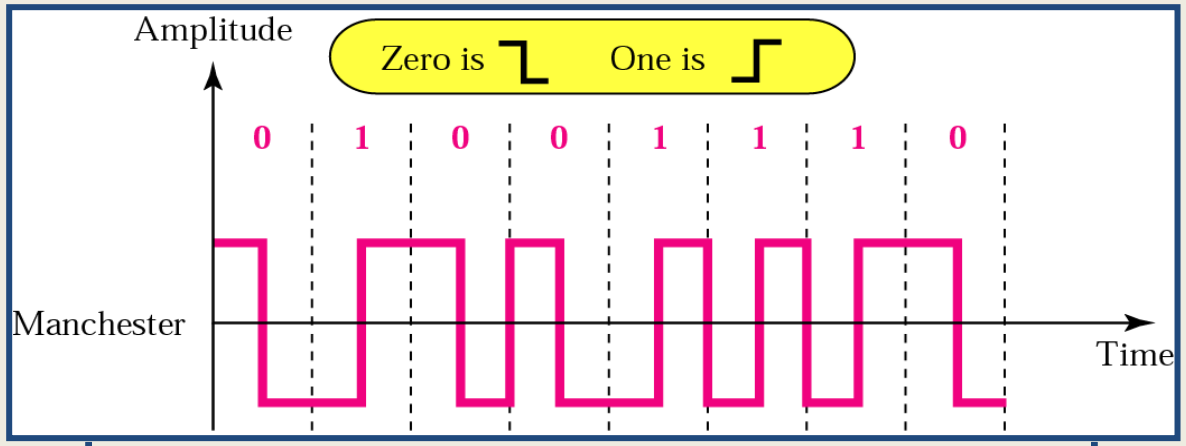
Three generations of Ethernet







Manchester encoding

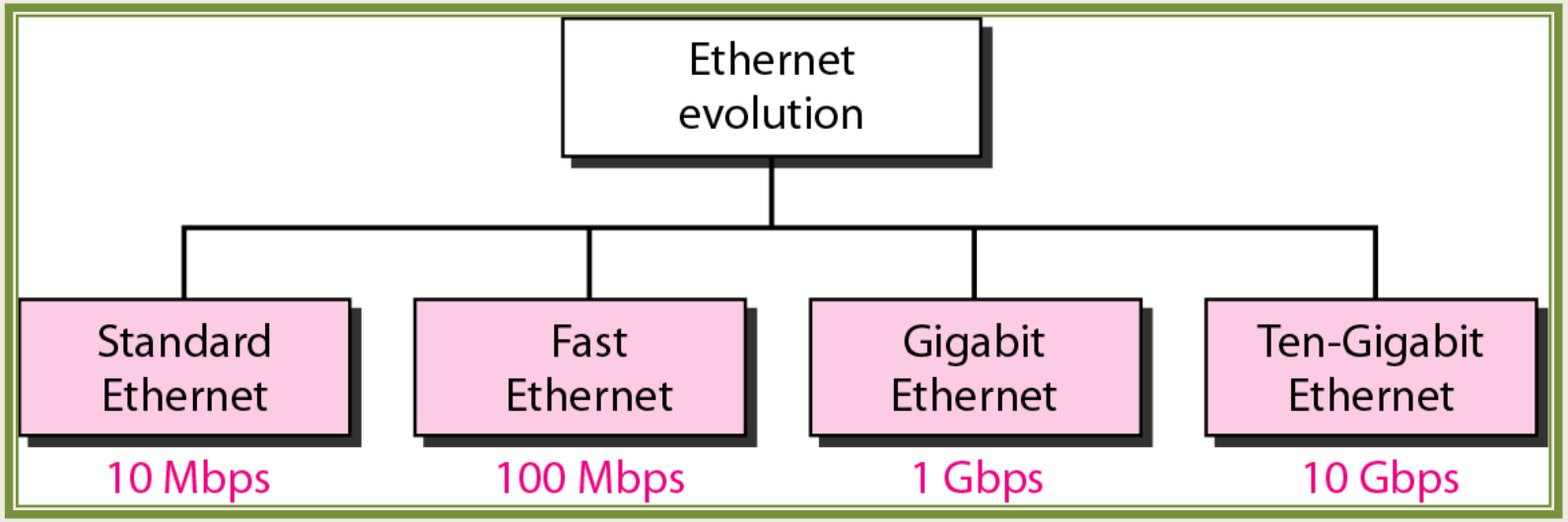


Four generations

- Standard Ethernet (10 Mbps)
- Fast Ethernet (100 Mbps)
- Gigabit Ethernet (1 Gbps)
- 10 Gigabit Ethernet (10 Gbps)



Ethernet evolution through four generations



Characteristics of the Standard Ethernet

Connectionless and Unreliable Service

- Ethernet provides a connectionless service, which means each frame sent is independent of the previous or next frame.
- Ethernet has no connection establishment or connection termination phases.
- The sender sends a frame whenever it has it; the receiver may or may not be ready for it.
- The sender may overwhelm the receiver with frames, which may result in dropping frames.

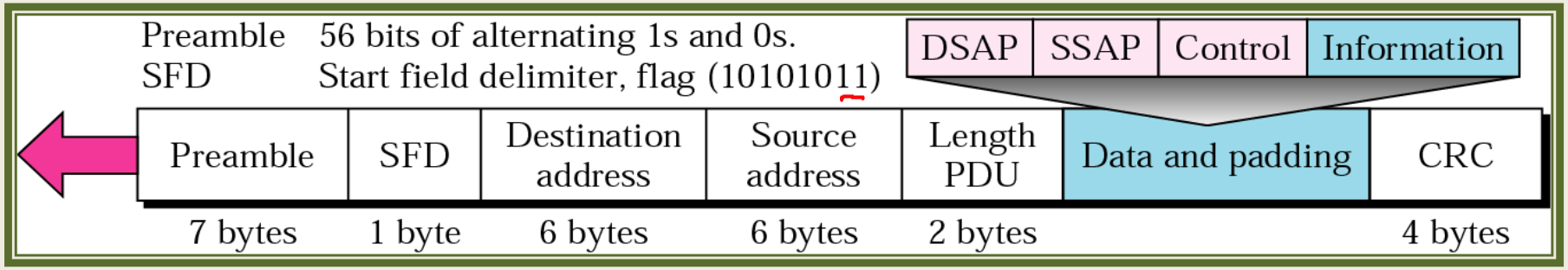
Contd...

- If a frame drops, the sender will not know about it.
- If the transport layer is also a connectionless protocol, such as UDP, the frame is lost and salvation may only come from the application layer.
- If the transport layer is TCP, the sender TCP does not receive acknowledgment for its segment and sends it again.
- Ethernet is also unreliable like IP and UDP.

Contd...

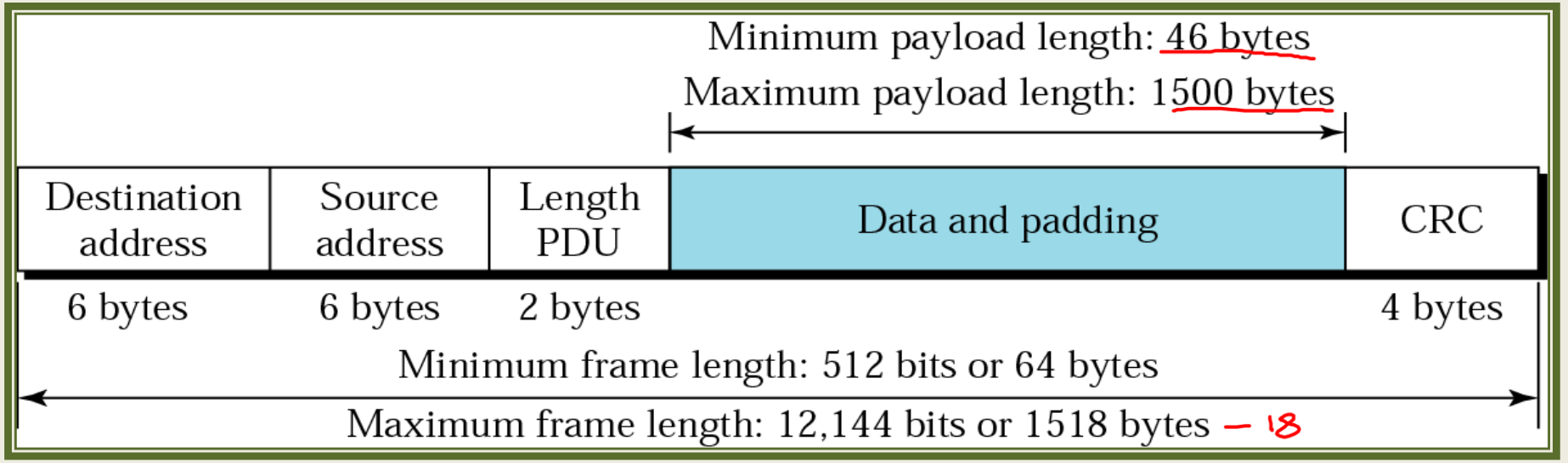
- If a frame is corrupted during transmission and the receiver finds out about the corruption, which has a high level of probability of happening because of the CRC-32, the receiver drops the frame silently.
- It is the duty of high-level protocols to find out about it.

802.3 MAC frame



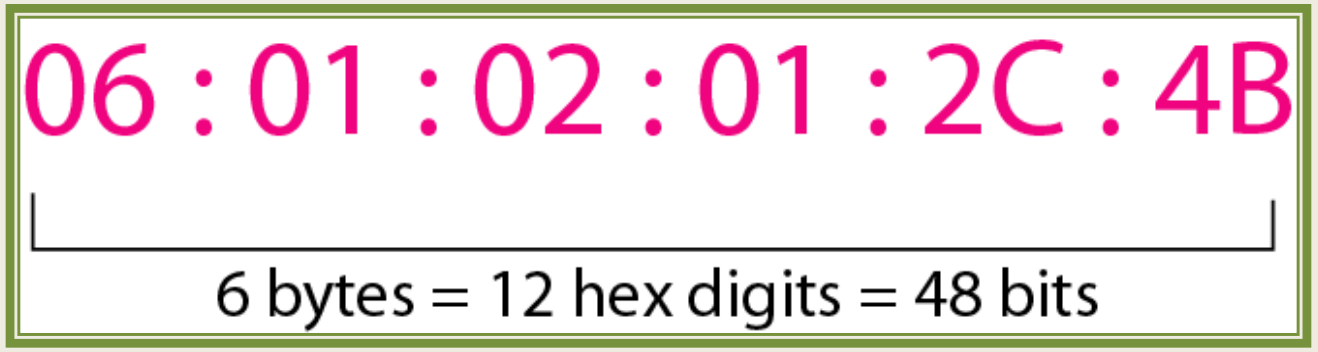
- ***Preamble:*** Provides an alert & a timing pulse
- ***SFD:*** Signals the beginning of the frame,& also warns the stations that this is the last the chance for synchronization. Last 2 bits alerts the receiver that next field is the destination address
- ***DA & SA*** : Physical address of receiver & sender
- ***Length or type:*** Define the upper layer protocol using the MAC frame IEEE use as length field to define the number of bytes in the data field

Minimum and maximum length

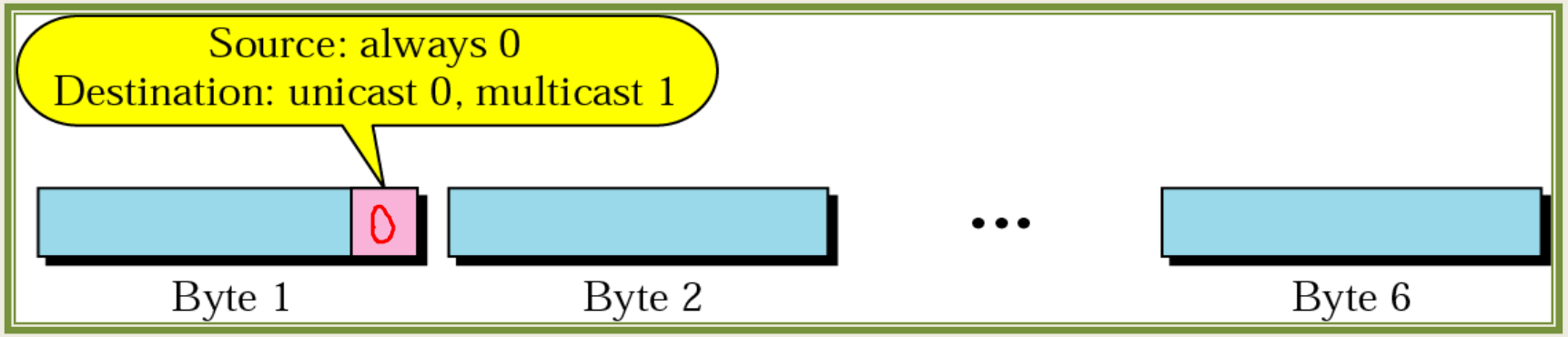


- *If 18 bytes of header and trailer(6 SA + 6 DA + 2 Length + 4 CRC)*
- *Then Minimum length of data from the upper layer is 64-18= 46 bytes*
- *If the upper-layer packet is less than 46 bytes, padding is added to make up the difference*

Example of an Ethernet address in hexadecimal notation



Unicast and multicast addressed



- Source address is always a unicast address
- Destination address can be unicast, multicast ,or broadcast
- LSB of the 1st byte defines the type of address
- Broadcast DA is a special case of the multicast address in which all bits are 1s

Contd...

- The least significant bit of the first byte defines the type of address. If the bit is 0, the address is unicast; otherwise, it is multicast.
- The broadcast destination address is a special case of the multicast address in which all bits are 1s.

Contd...

- Access method: standard ethernet uses 1-persistent CSMA/CD
- SLOT TIME: round trip time + time required to send the jam sequence
- Defined in bits,time required to send 512 bits
- Depends on the data rate
- For 10-Mbps ethernet it is 51.2 μs

Example

Define the type of the following destination addresses:

a. 4A:30:10:21:10:1A b. 47:20:1B:2E:08:EE c. FF:FF:FF:FF:FF:FF

Solution

To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are F's, the address is broadcast.

a. This is a unicast address because A in binary is 1010.

b. This is a multicast address because 7 in binary is 0111.

c. This is a broadcast address because all digits are F's.

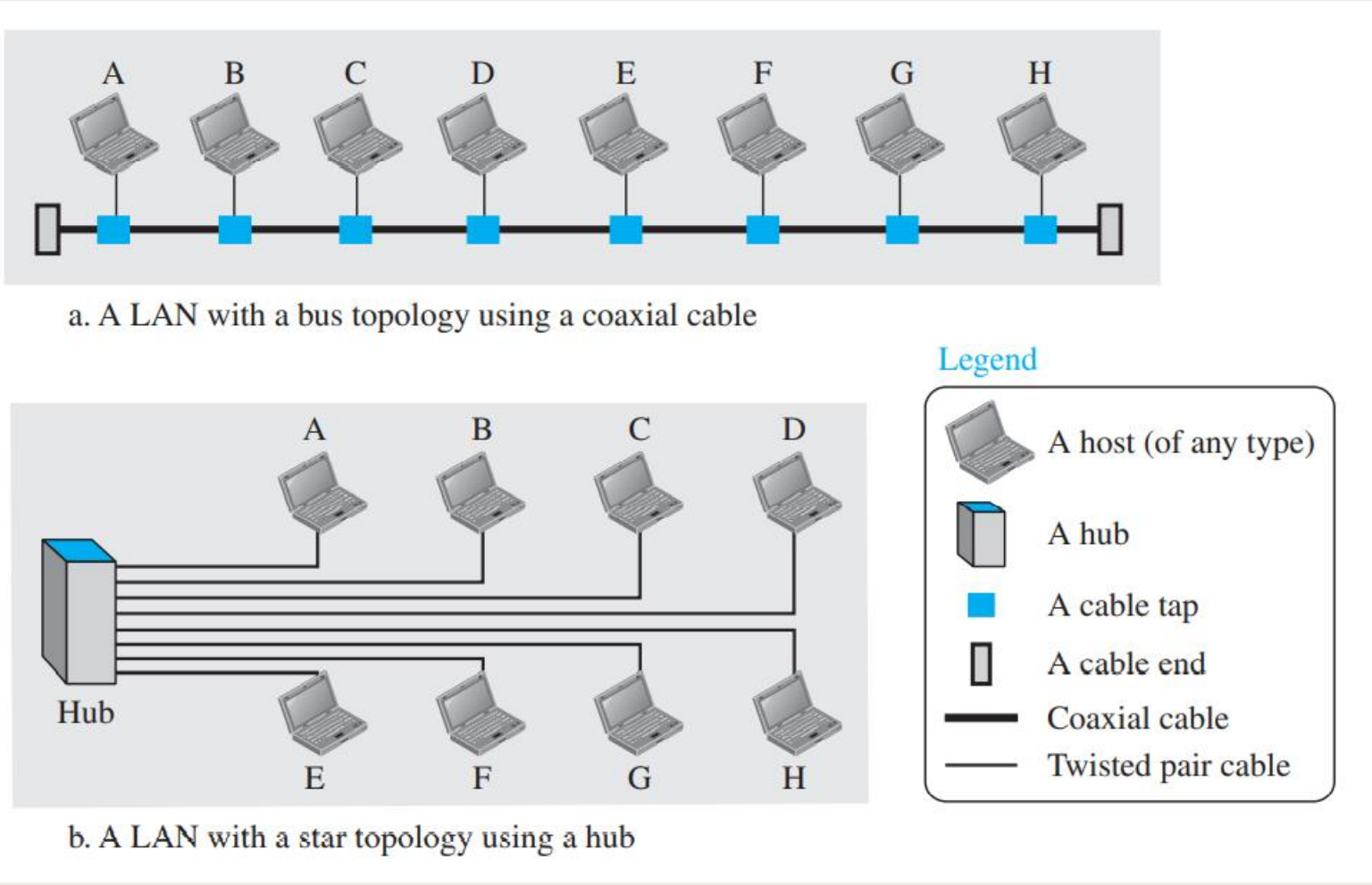
Distinguish Between Unicast, Multicast, and Broadcast Transmission

- Transmission in the standard Ethernet is always broadcast, no matter if the intention is unicast, multicast, or broadcast.
- In the bus topology, when station A sends a frame to station B, all stations will receive it.
- In the star topology, when station A sends a frame to station B, the hub will receive it.
- Since the hub is a passive element, it does not check the destination address of the frame; it regenerates the bits (if they have been weakened) and sends them to all stations except station A.

Contd...

- In a unicast transmission, all stations will receive the frame, the intended recipient keeps and handles the frame; the rest discard it.
- In a multicast transmission, all stations will receive the frame, the stations that are members of the group keep and handle it; the rest discard it.
- In a broadcast transmission, all stations (except the sender) will receive the frame and all stations (except the sender) keep and handle it.

Implementation of standard Ethernet



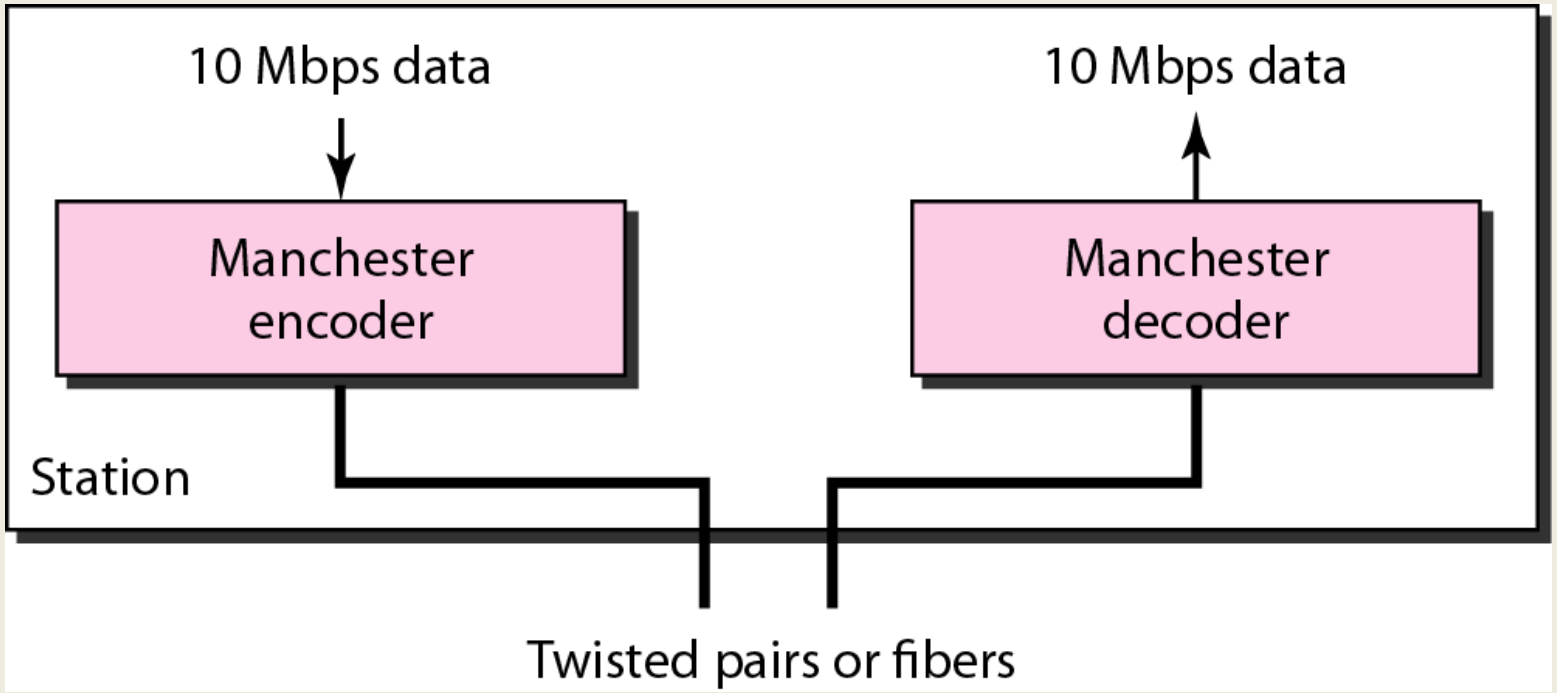
a. A LAN with a bus topology using a coaxial cable

b. A LAN with a star topology using a hub

Efficiency of Standard Ethernet

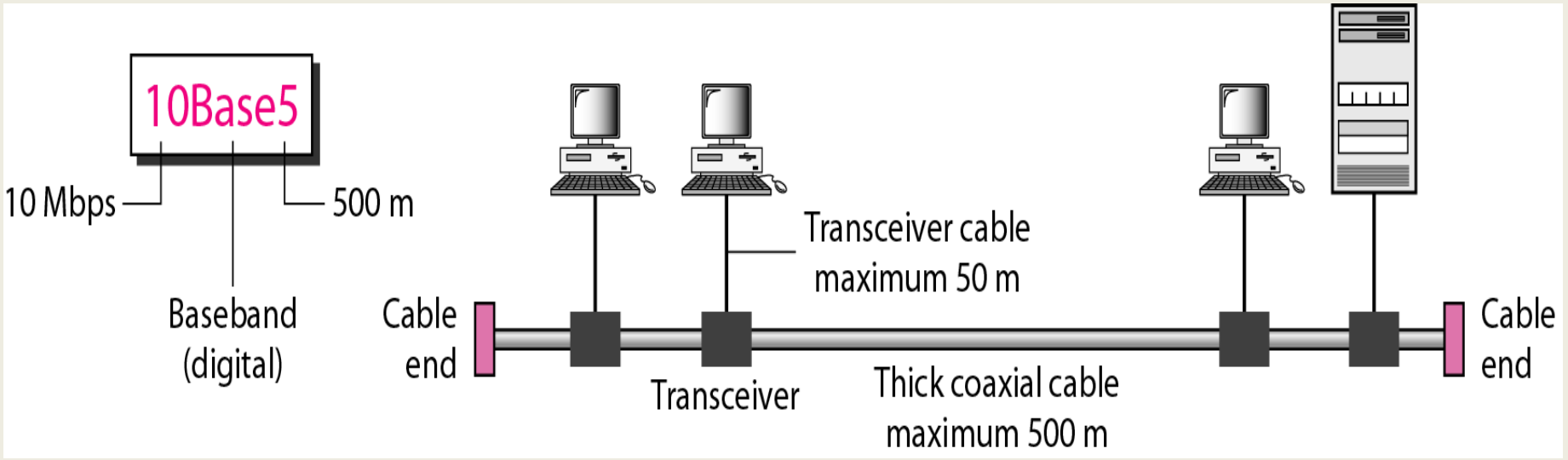
- The efficiency of the Ethernet is defined as the ratio of the time used by a station to send data to the time the medium is occupied by this station
- Efficiency = 1 / (1 + 6.4 × a)
- "a" is the number of frames that can fit on the medium.
- It can be calculated as a = (propagation delay)/(transmission delay) because the transmission delay is the time it takes a frame of average size to be sent out and the propagation delay is the time it takes to reach the end of the medium

Encoding in a Standard Ethernet implementation

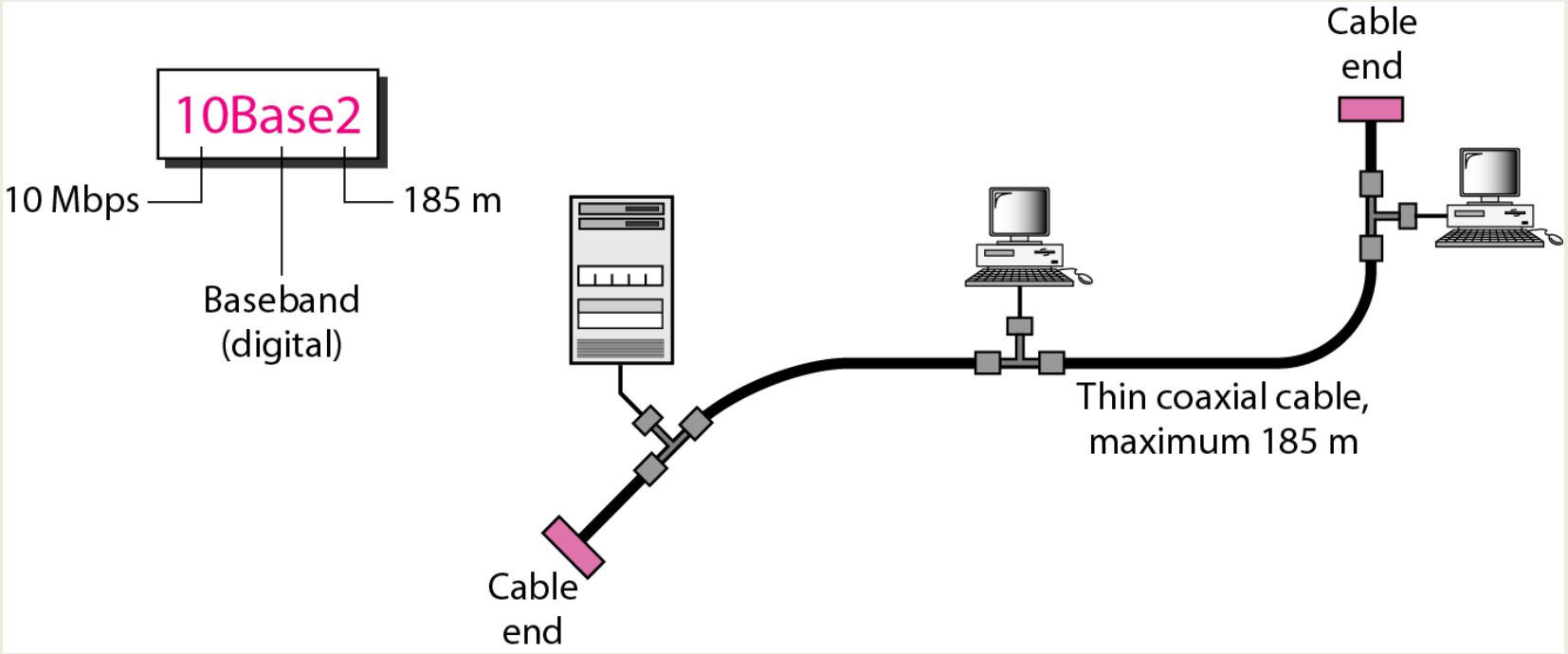


10Base5 implementation

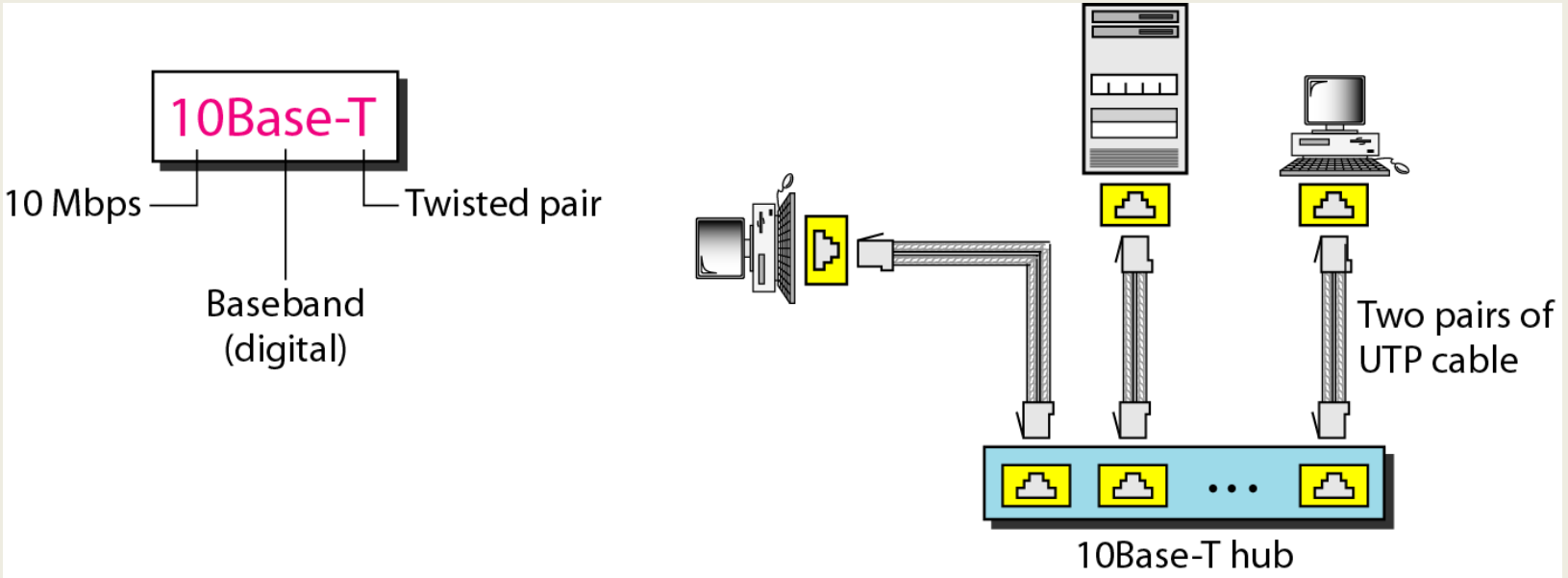
Bassband → modulated
Pass band →



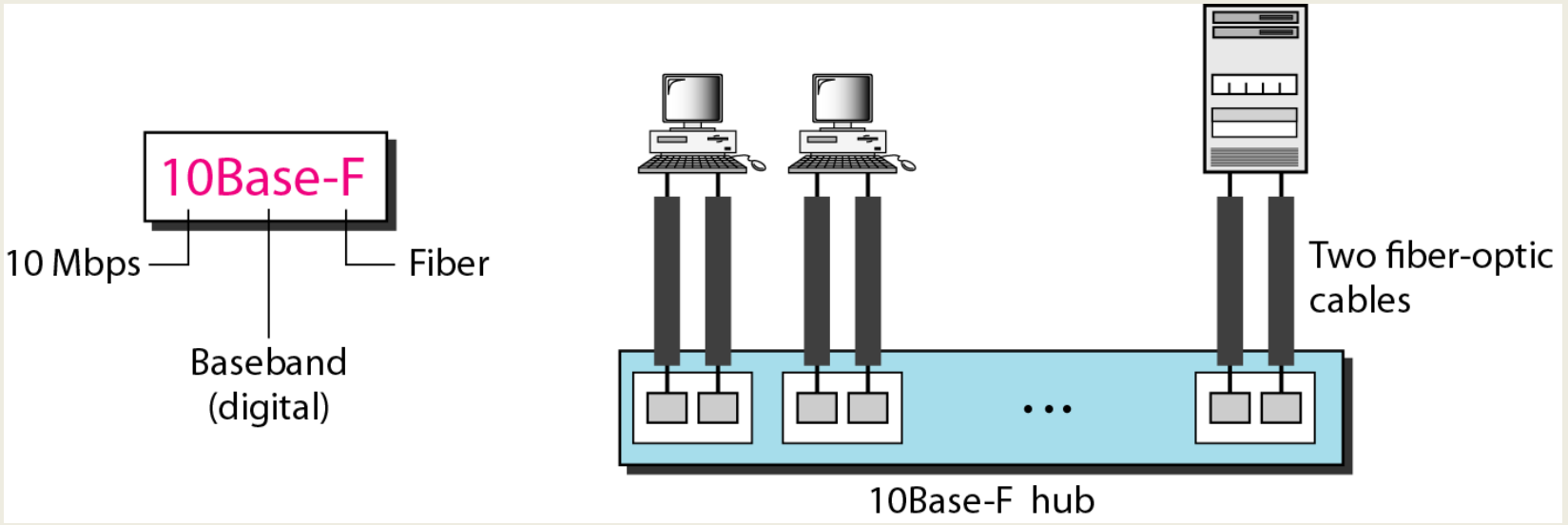
10Base2 implementation



10Base-T implementation



10Base-F implementation



Summary of Standard Ethernet implementations

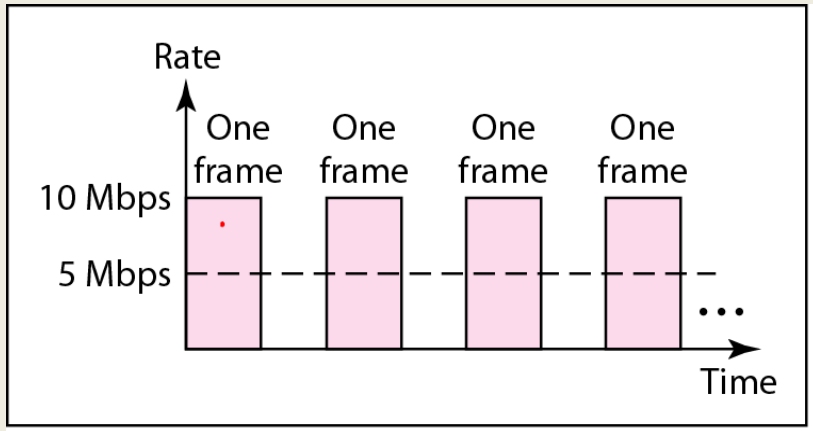
Characteristics	*10Base5*	*10Base2*	*10Base-T*	*10Base-F*
Media	Thick coaxial cable	Thin coaxial cable	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m
Line encoding	Manchester	Manchester	Manchester	Manchester

CHANGES IN THE STANDARD

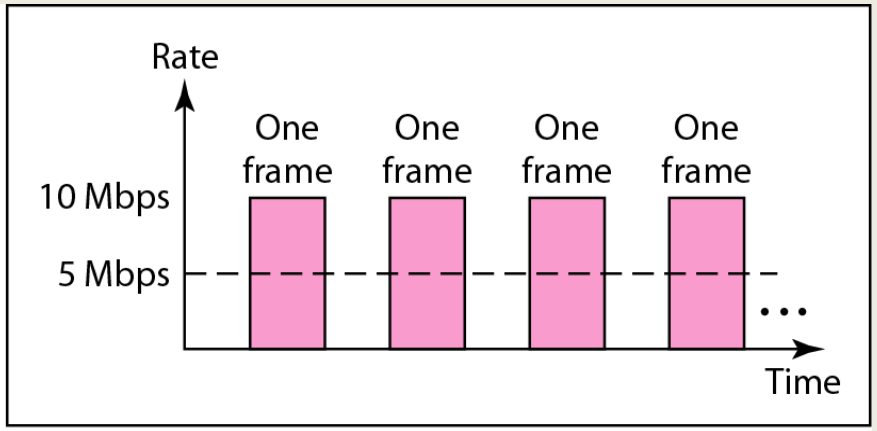
The 10-Mbps Standard Ethernet has gone through several changes before moving to the higher data rates. These changes actually opened the road to the evolution of the Ethernet to become compatible with other high-data-rate LANs.

- **Bridged Ethernet**
- **Switched Ethernet**
- **Full-Duplex Ethernet**

Sharing bandwidth

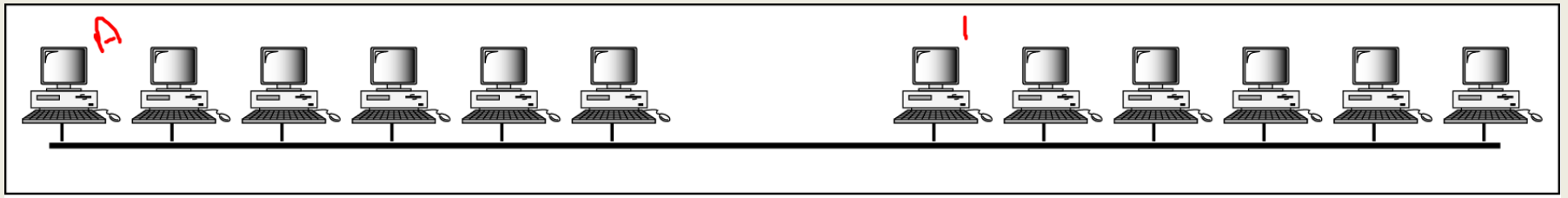


a. First station

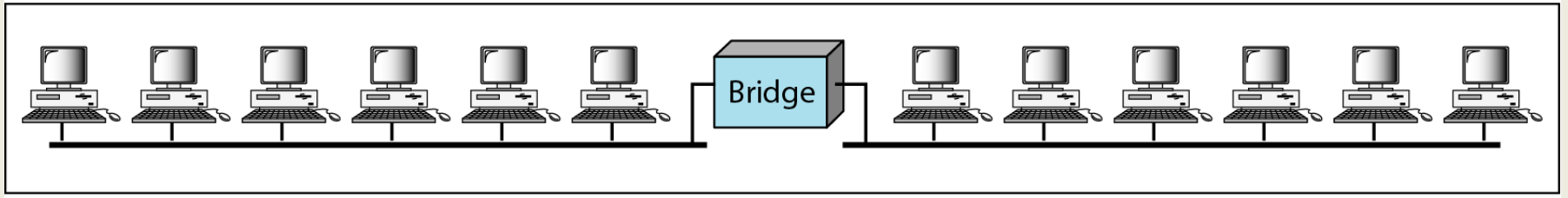


b. Second station

A network with and without a bridge



a. Without bridging



b. With bridging

Contd...

- In an unbridged Ethernet network, the total capacity (10 Mbps) is shared among all stations with a frame to send; the stations share the bandwidth of the network.
- If only one station has frames to send, it benefits from the total capacity (10 Mbps).
- For example, if two stations have a lot of frames to send, they probably alternate in usage.
- When one station is sending, the other one refrains from sending. Each station on average sends at a rate of 5 Mbps.

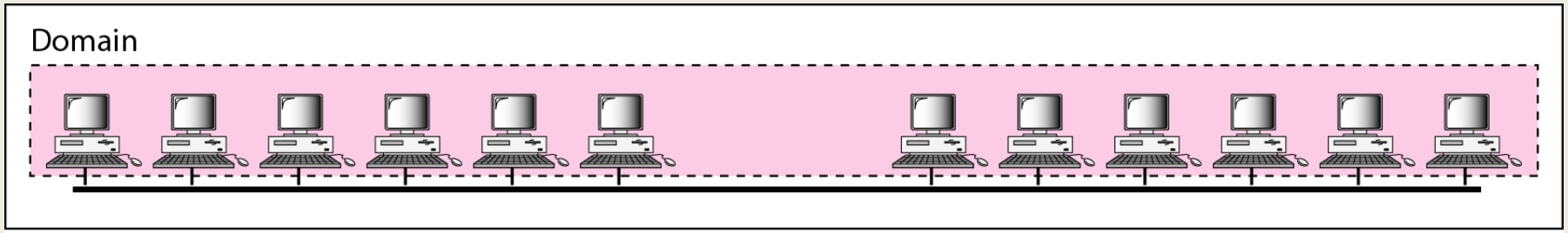
Contd...

- A bridge divides the network into two or more networks. Bandwidth wise, each network is independent.
- For example, a network with 12 stations is divided into two networks, each with 6 stations. Each network has a capacity of 10 Mbps.
- The 10-Mbps capacity in each segment is now shared between 6 stations station in each segment), not 12 stations.
- In a network with a heavy load, each station theoretically is offered 10/7 Mbps instead of 10/12 Mbps.

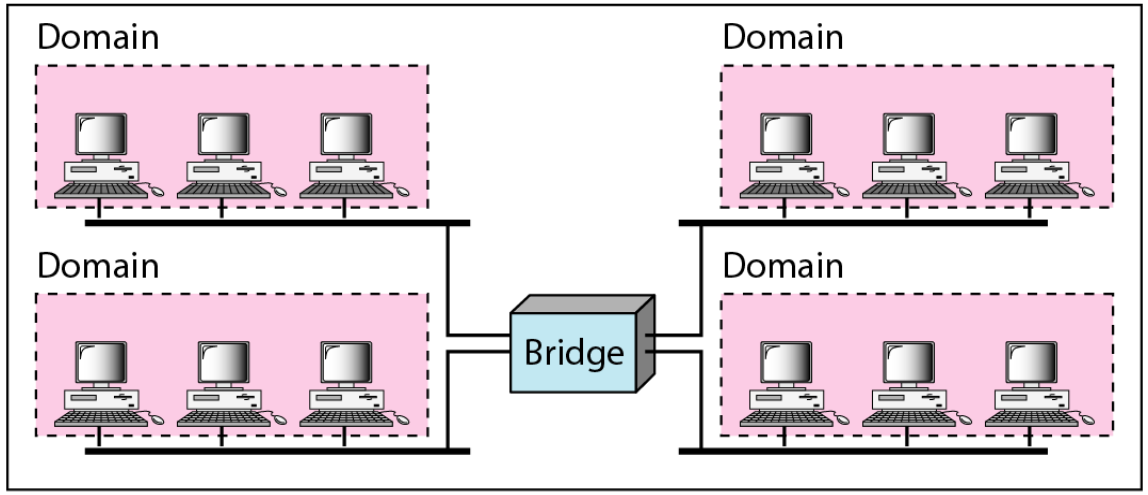
Separating Collision Domains

- Another advantage of a bridge is the separation of the collision domain
- The collision domain becomes much smaller and the probability of collision is reduced tremendously.
- Without bridging, 12 stations contend for access to the medium; with bridging only 3 stations contend for access to the medium

Collision domains in an unbridged network and a bridged network



a. Without bridging

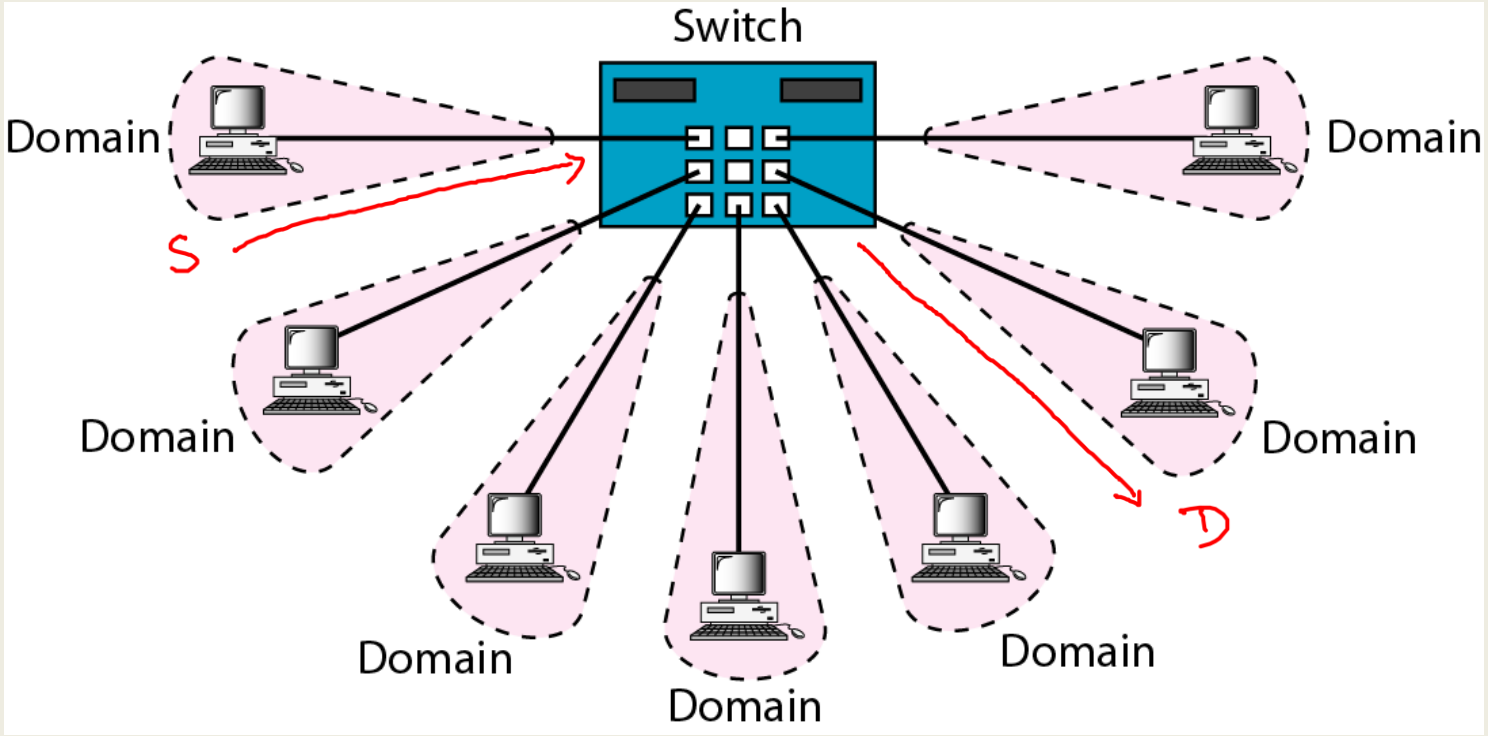


b. With bridging

Switched Ethernet

- A layer-2 switch is an N-port bridge with additional sophistication that allows faster handling of the packets.
- Evolution from a bridged Ethernet to a switched Ethernet was a big step that opened the way to faster Ethernet

Switched Ethernet



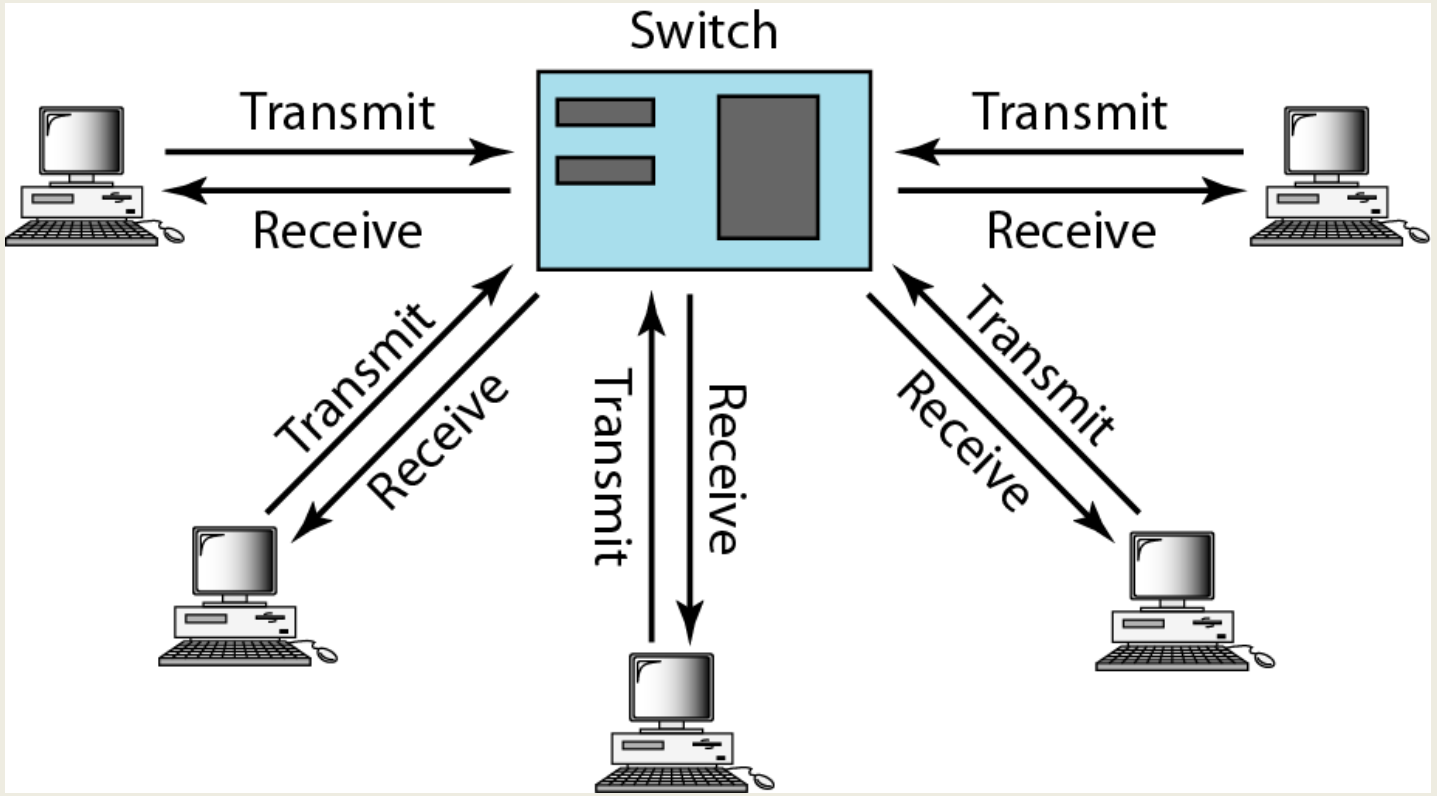
Full-Duplex Ethernet

- One of the limitations of 10Base5 and 10Base2 is that communication is half-duplex (10Base-T is always full-duplex); a station can either send or receive.
- The full-duplex mode increases the capacity of each domain from 10 to 20 Mbps.
- The configuration uses two links: one to transmit and one to receive.
- no need for the CSMA/CD method

Contd...

- Each station or switch can send and receive independently without worrying about collision.
- Each link is a point-to-point dedicated path between the station and the switch.
- The carrier sensing and collision detection functionalities of the MAC sublayer can be turned off.

Full-duplex switched Ethernet



MAC Control Layer

- To provide for flow and error control in full-duplex switched Ethernet, a new sublayer, called the MAC control, is added between the LLC sublayer and the MAC sublayer.

FAST ETHERNET

Fast Ethernet was designed to compete with LAN protocols such as FDDI or Fiber Channel. IEEE created Fast Ethernet under the name 802.3u. Fast Ethernet is backward-compatible with Standard Ethernet, but it can transmit data 10 times faster at a rate of 100 Mbps.

MAC Sublayer
Physical Layer

Goals of Fast Ethernet

1. Upgrade the data rate to 100 Mbps.
2. Make it compatible with Standard Ethernet.
3. Keep the same 48-bit address.
4. Keep the same frame format.

Access Method

- The first solution was to totally drop the bus topology and use a passive hub and star topology but make the maximum size of the network 250 meters instead of 2500 meters as in the Standard Ethernet.
- The second solution is to use a link-layer switch (discussed later in the chapter) with a buffer to store frames and a full-duplex connection to each host to make the transmission medium private for each host.
- There is no need for CSMA/CD because the hosts are not competing with each other.

Contd...

- The link-layer switch receives a frame from a source host and stores it in the buffer (queue) waiting for processing.
- It then checks the destination address and sends the frame out of the corresponding interface.
- Since the connection to the switch is full-duplex, the destination address can even send a frame to another station at the same time that it is receiving a frame.
- The shared medium is changed to many point-to-point media, and there is no need for contention.

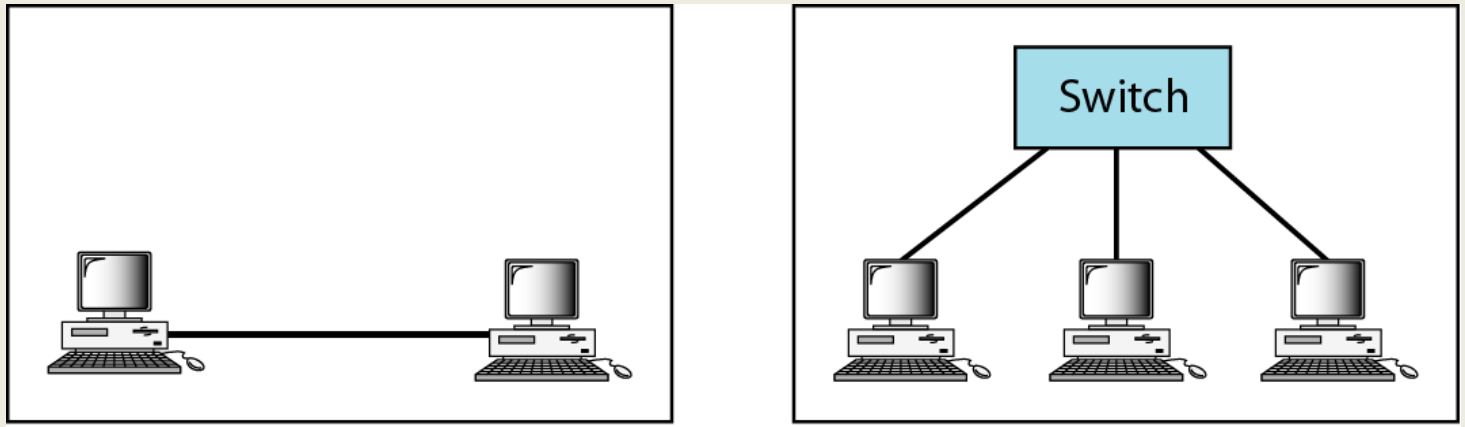
Auto negotiation

- A new feature added to Fast Ethernet is called auto negotiation. It allows a station or a hub a range of capabilities.
- Auto negotiation allows two devices to negotiate the mode or data rate of operation.
- It was designed particularly to allow incompatible devices to connect to one another.
- For example, a device with a maximum data rate of 10 Mbps can communicate with a device with a 100 Mbps data rate (but which can work at a lower rate).

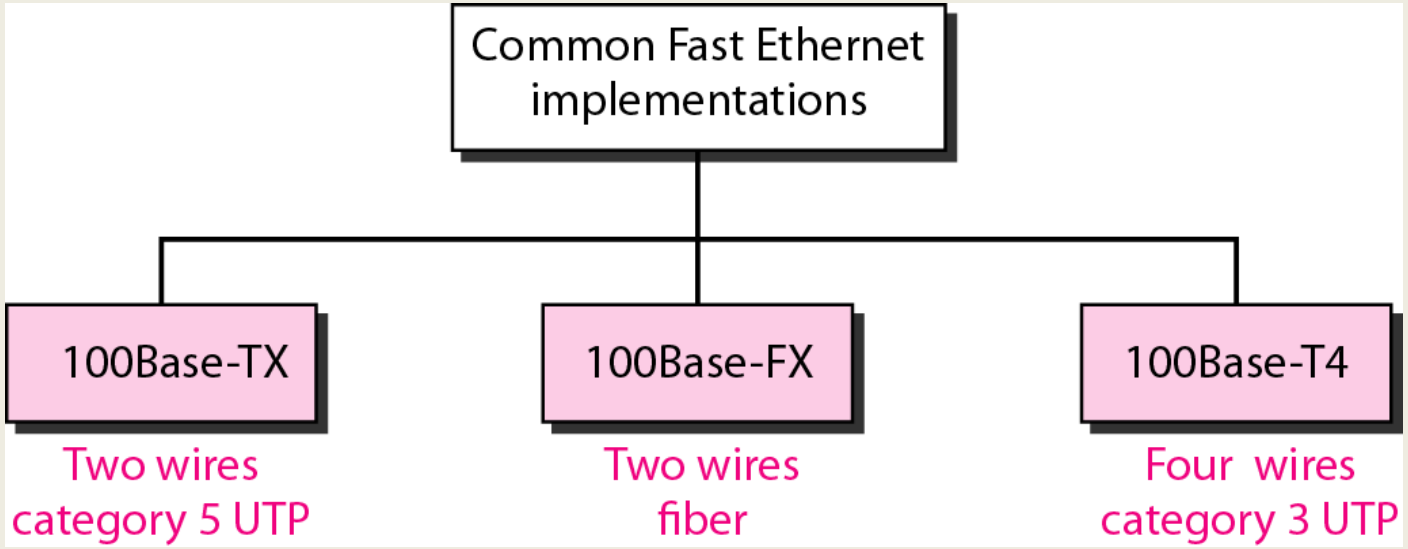
Goal of auto-negotiation

- To allow incompatible devices to connect to one another.
- To allow one device to have multiple capabilities.
- To allow a station to check a hub's capabilities.

Fast Ethernet topology



Fast Ethernet implementations



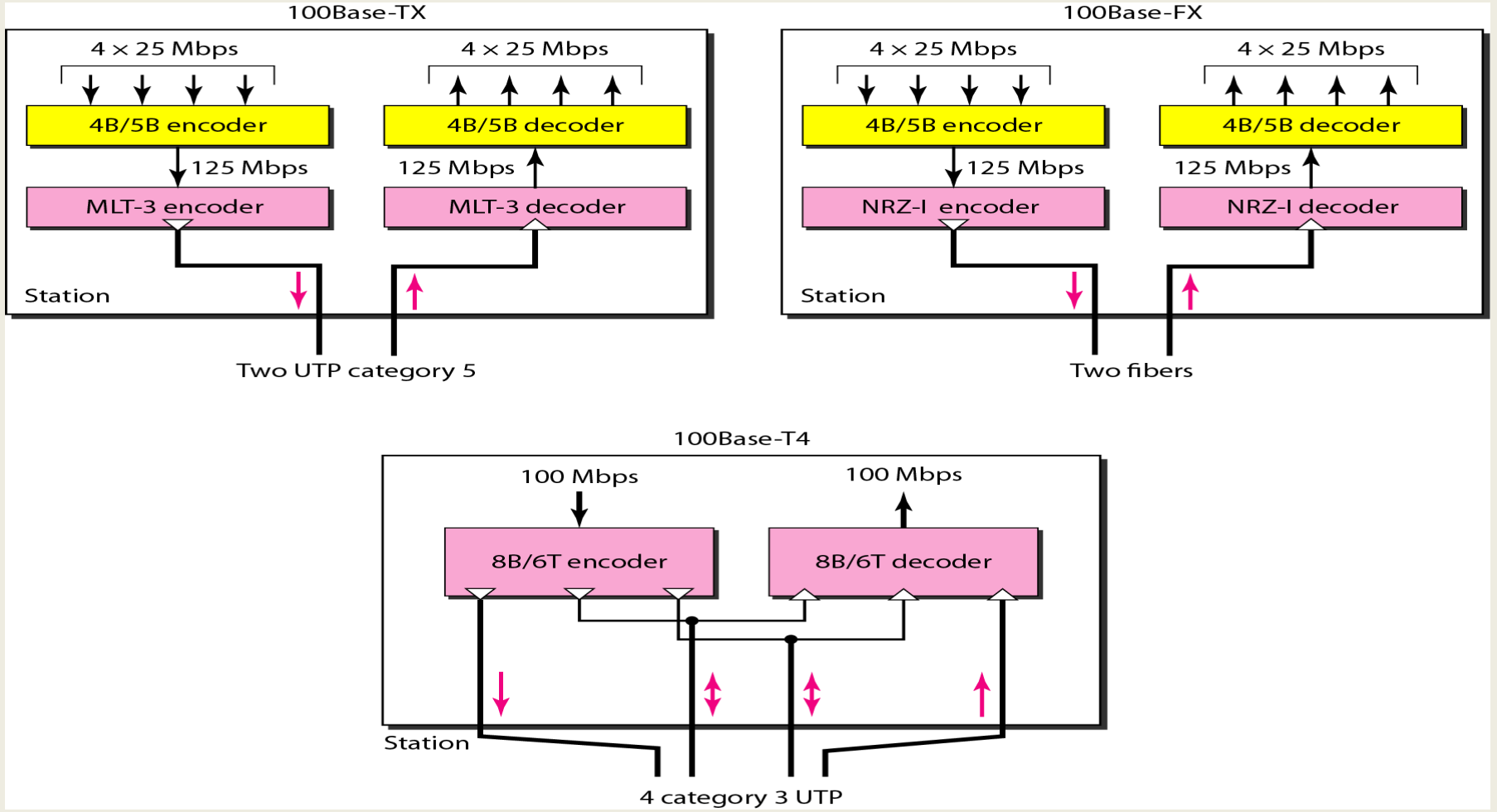
Contd...

- 100Base-TX
  - uses two pairs of twisted-pair cable (either category 5 UTP or STP).
  - For this implementation, the MLT-3 scheme was selected since it has good bandwidth performance
- 100Base-FX
  - uses two pairs of fiber-optic cables.
  - Optical fiber can easily handle high bandwidth requirements by using simple encoding schemes.
  - The designers of 100Base-FX selected the NRZ-I encoding scheme for this implementation

Contd...

- A 100Base-TX network can provide a data rate of 100 Mbps, but it requires the use of category 5 UTP or STP cable
- This is not cost-efficient for buildings that have already been wired for voice-grade twisted-pair (category 3).
- A new standard, called 100Base-T4, was designed to use category 3 or higher UTP.
- The implementation uses four pairs of UTP for transmitting 100 Mbps

Encoding for Fast Ethernet implementation



Summary of Fast Ethernet implementations

Characteristics	*100Base-TX*	*100Base-FX*	*100Base-T4*
Media	Cat 5 UTP or STP	Fiber	Cat 4 UTP
Number of wires	2	2	4
Maximum length	100 m	100 m	100 m
Block encoding	4B/5B	4B/5B	-
Line encoding	MLT-3	NRZ-I	8B/6T

Ethernet, Token bus & Token ring

GIGABIT ETHERNET

The need for an even higher data rate resulted in the design of the Gigabit Ethernet protocol (1000 Mbps). The IEEE committee calls the standard 802.3z.

- **MAC Sublayer**
- **Physical Layer**
- **Ten-Gigabit Ethernet**

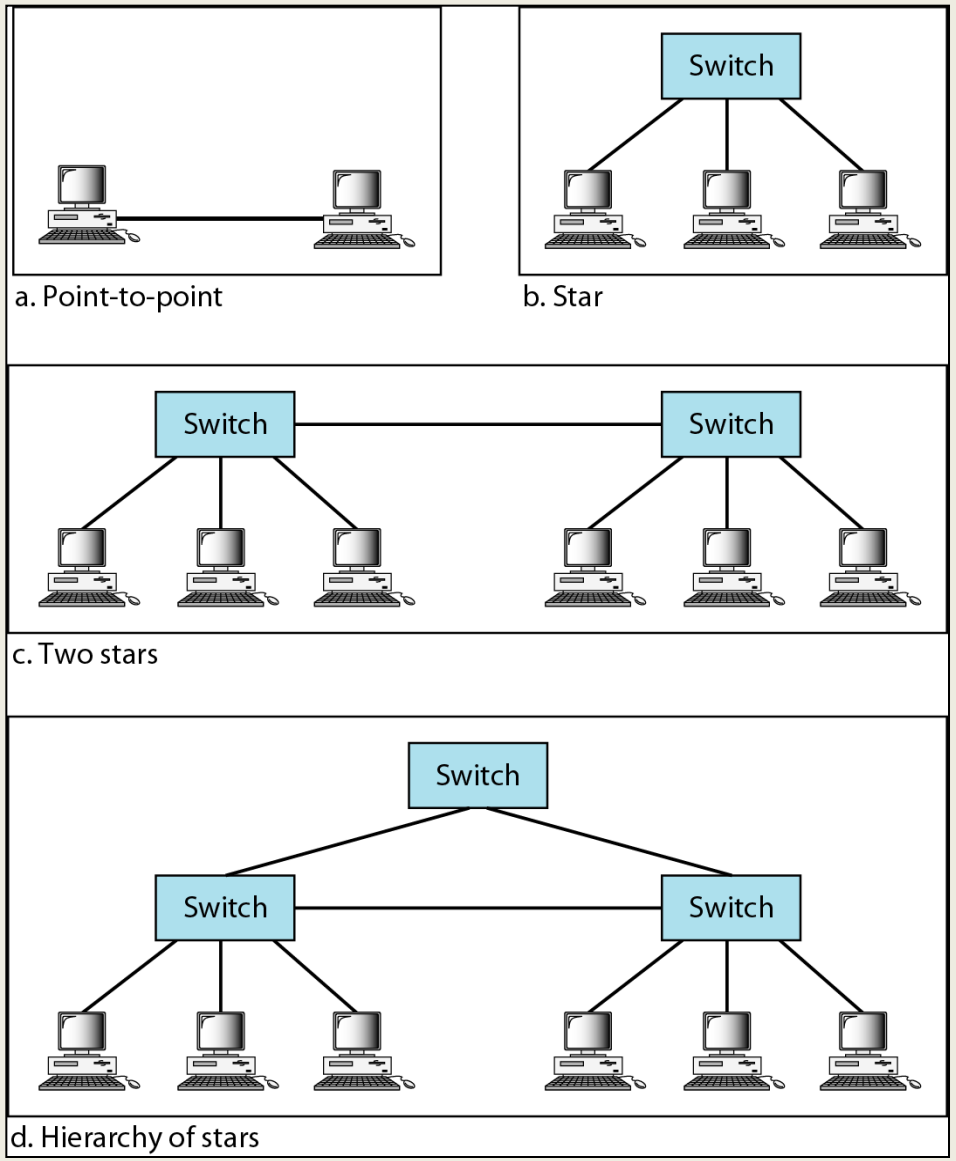
Note

In the full-duplex mode of Gigabit Ethernet, there is no collision; the maximum length of the cable is determined by the signal attenuation in the cable.

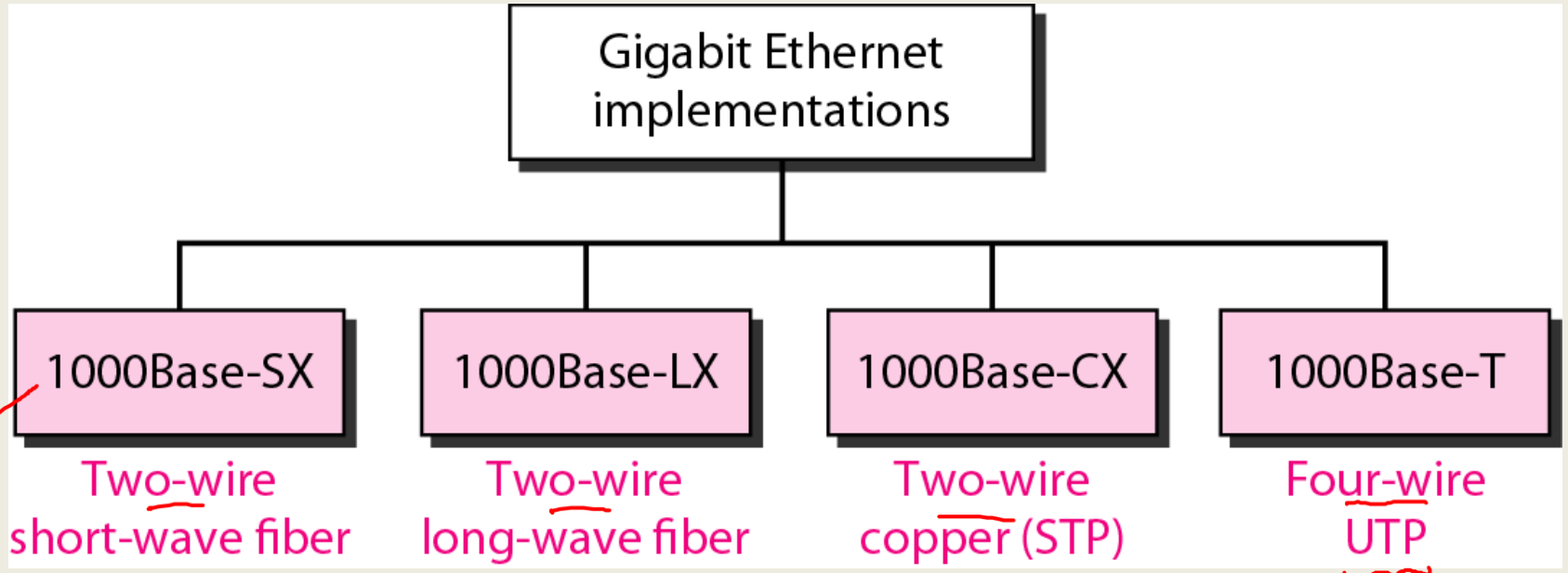
Goals of the Gigabit Ethernet design

- Upgrade the data rate to 1 Gbps.
- Make it compatible with Standard or Fast Ethernet.
- Use the same 48-bit address.
- Use the same frame format.
- Keep the same minimum and maximum frame lengths.
- Support auto negotiation as defined in Fast Ethernet.

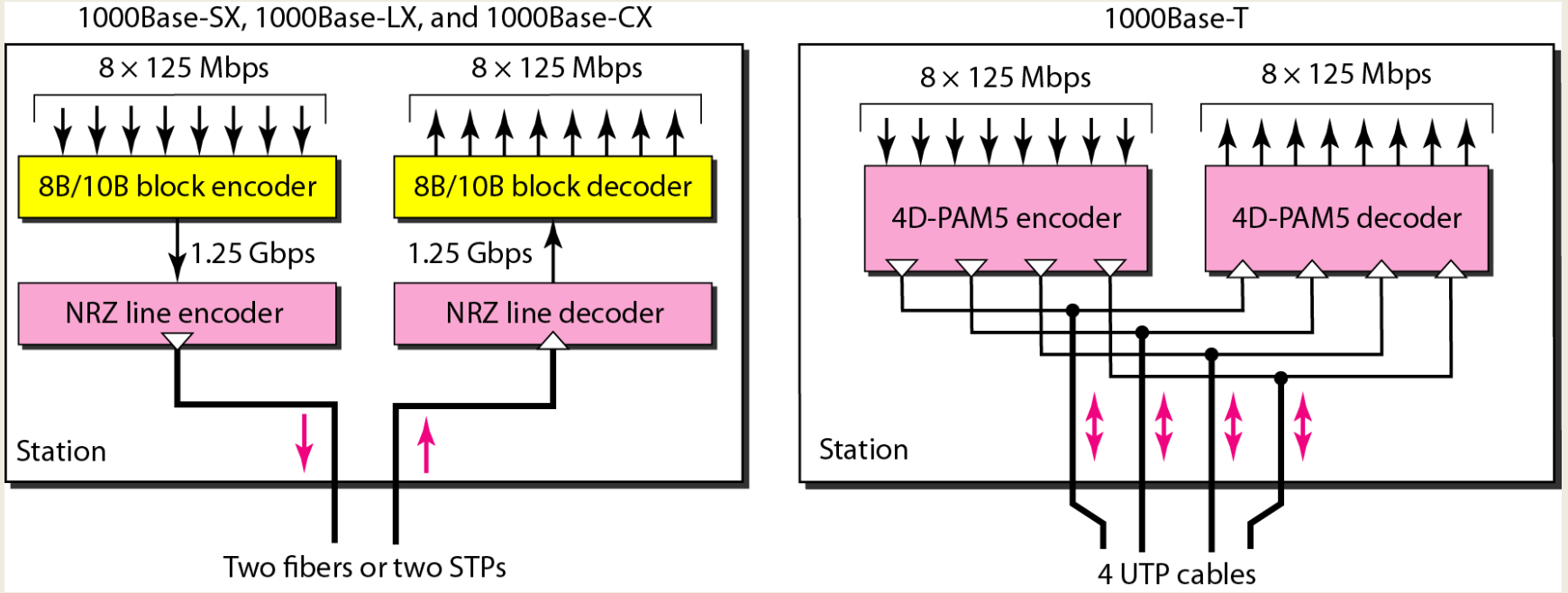
Topologies of Gigabit Ethernet



Gigabit Ethernet implementations



Encoding in Gigabit Ethernet implementations



Summary of Gigabit Ethernet implementations

Characteristics	*1000Base-SX*	*1000Base-LX*	*1000Base-CX*	*1000Base-T*
Media	Fiber short-wave	Fiber long-wave	STP	Cat 5 UTP
Number of wires	2	2	2	4
Maximum length	550 m	5000 m	25 m	100 m
Block encoding	8B/10B	8B/10B	8B/10B	
Line encoding	NRZ	NRZ	NRZ	4D-PAM5

Summary of Ten-Gigabit Ethernet implementations

Characteristics	*10GBase-S*	*10GBase-L*	*10GBase-E*
Media	Short-wave 850-nm multimode	Long-wave 1310-nm single mode	Extended 1550-mm single mode
Maximum length	300 m	10 km	40 km

Token Bus

IEEE 802.4 standard

IEEE 802.4 (Token Bus)

- Employs bus topology, no contention
- Uses single token passed around to all devices in order
- Device can only transmit when it has token
- Central to Manufacturing Automation Protocol (MAP) – connects robots and other machines on assembly line by a LAN
- Physically, it is a Bus network. Logically, it is a Ring network
- Stations are organized as a circular doubly-linked list

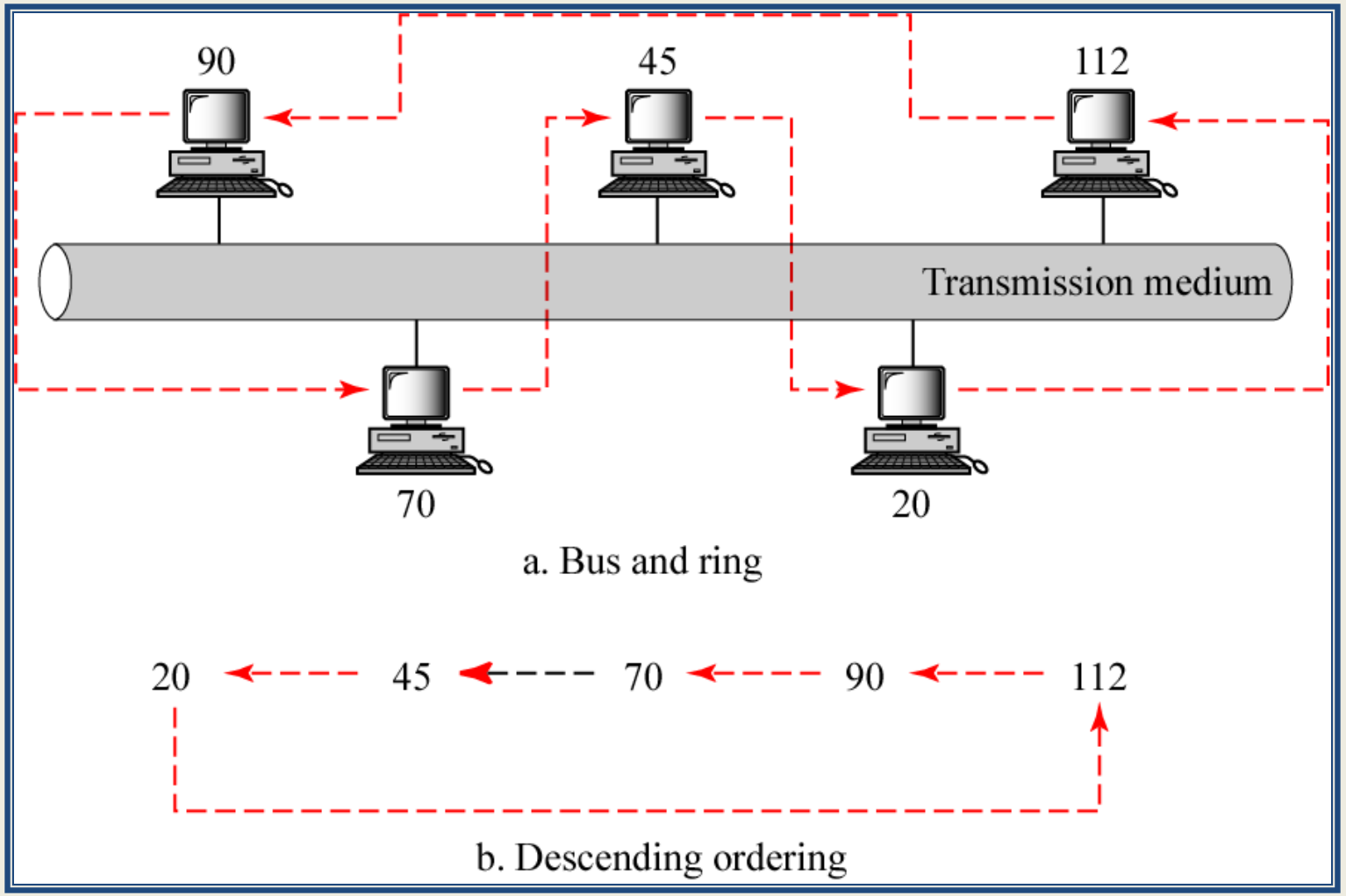
Contd...

- A distributed polling algorithm is used to avoid bus contention
- Token: Right of access
- Token Holder (The station receiving the token)
  - Transmit one or more MAC-frame
  - Poll other stations
  - Receive responses

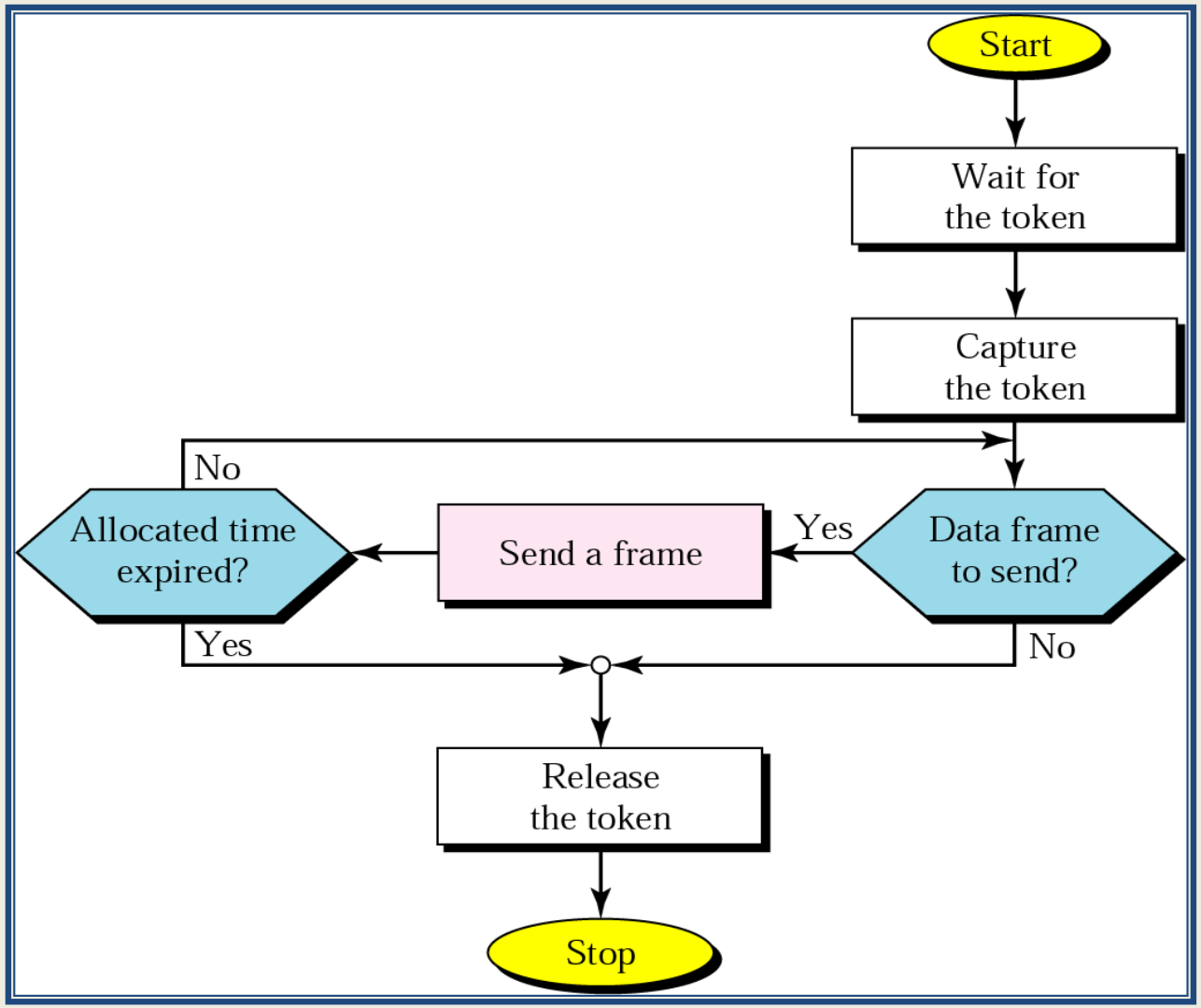
Token-Passing Bus Access Method

- Non-Token Holder
  - *Listen to the channel*
  - *Respond to a poll*
  - *Send Acknowledgement*
- Logical Ring Maintenance
  - *Ring Initialization*
  - *Addition to ring*
  - *Deletion from ring*
  - *Error Recovery*

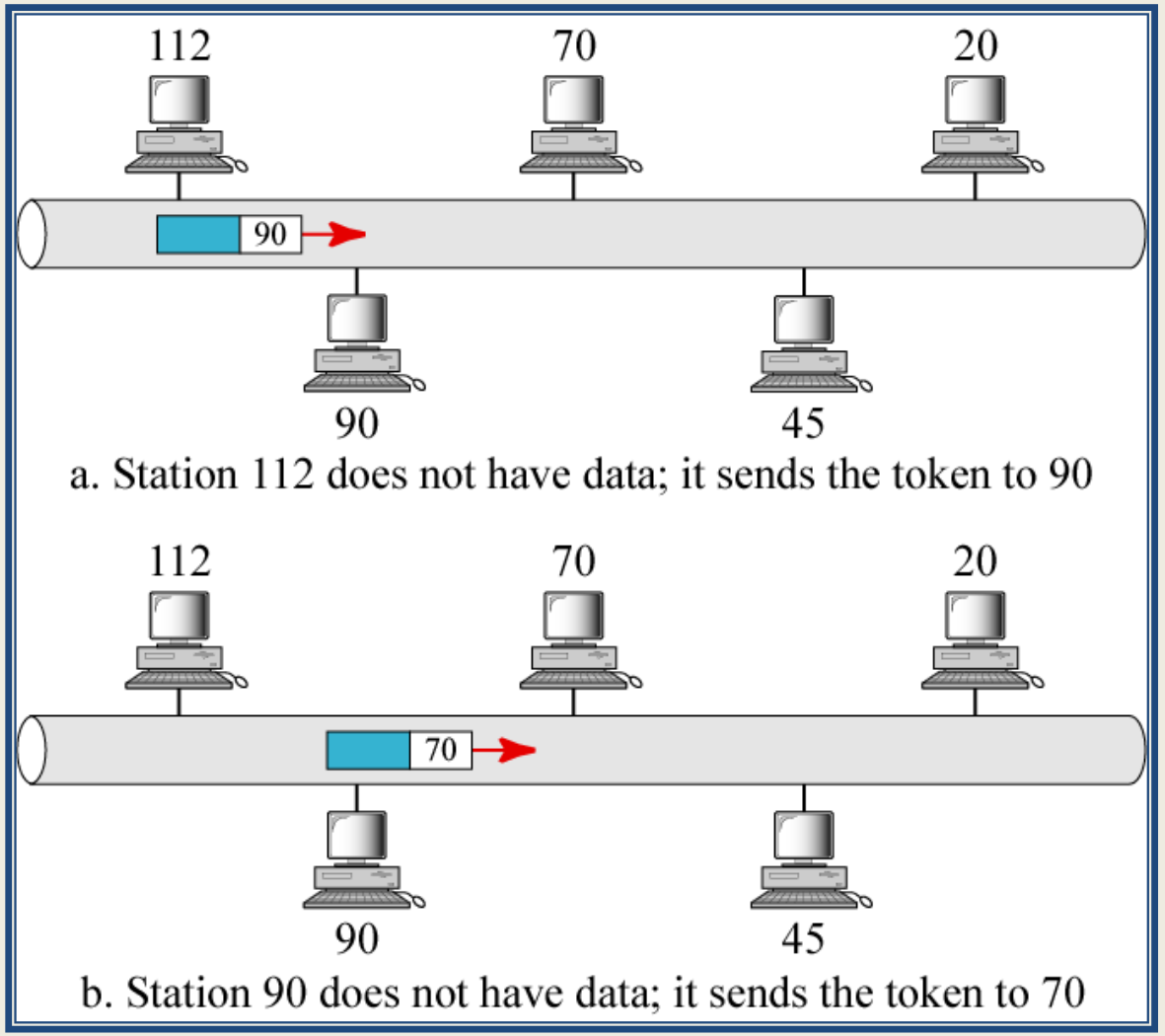
A Token Bus Network



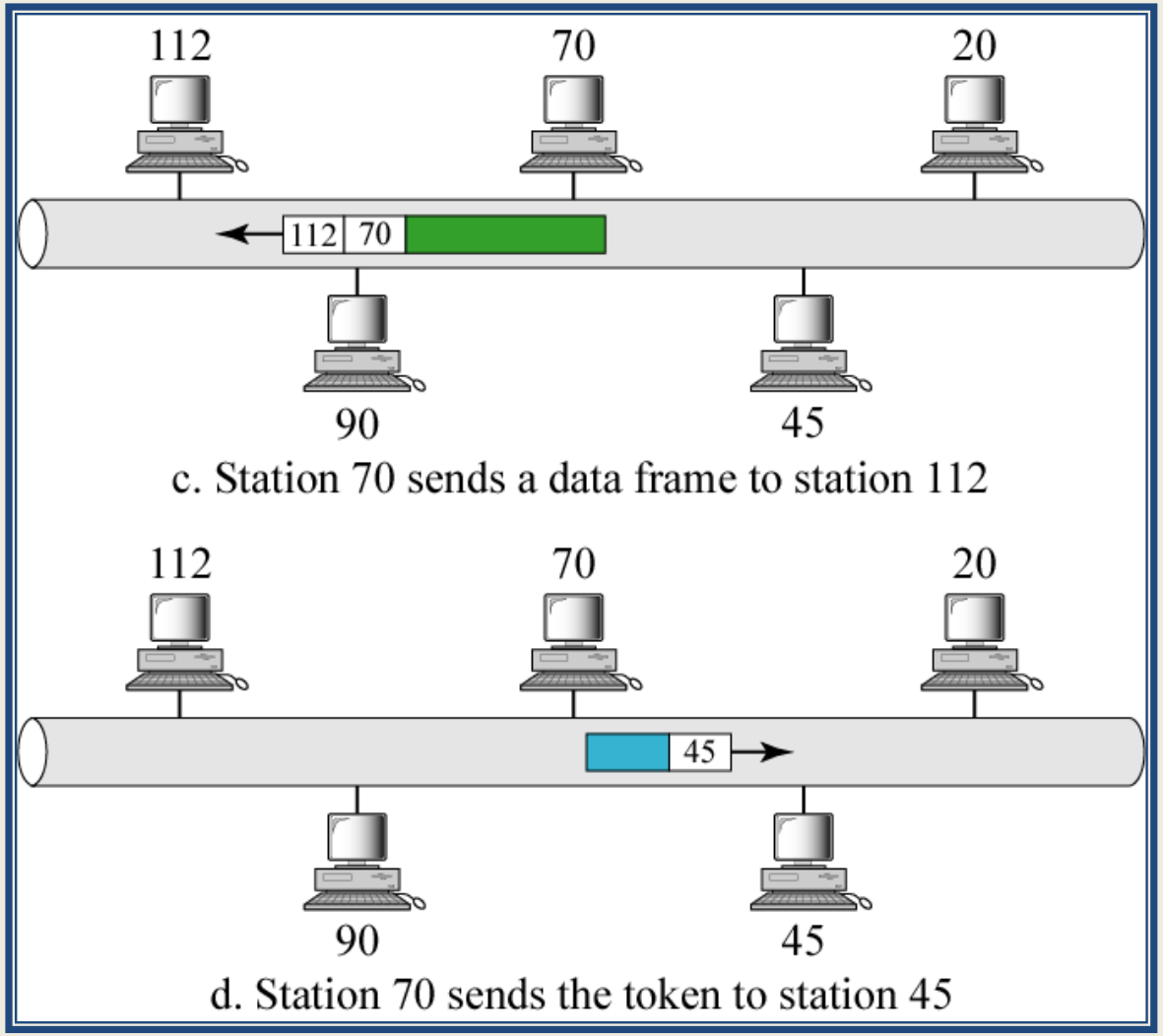
Token-passing procedure



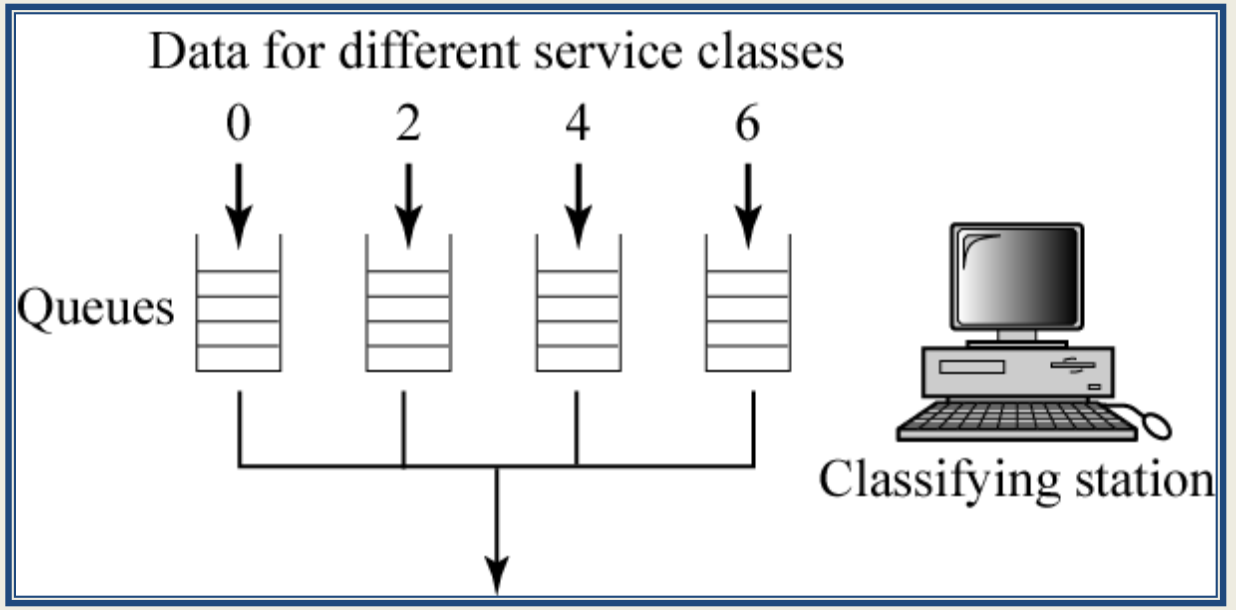
Contd...


112
70
20
90
90
45
a. Station 112 does not have data; it sends the token to 90
112
70
20
70
90
45
b. Station 90 does not have data; it sends the token to 70

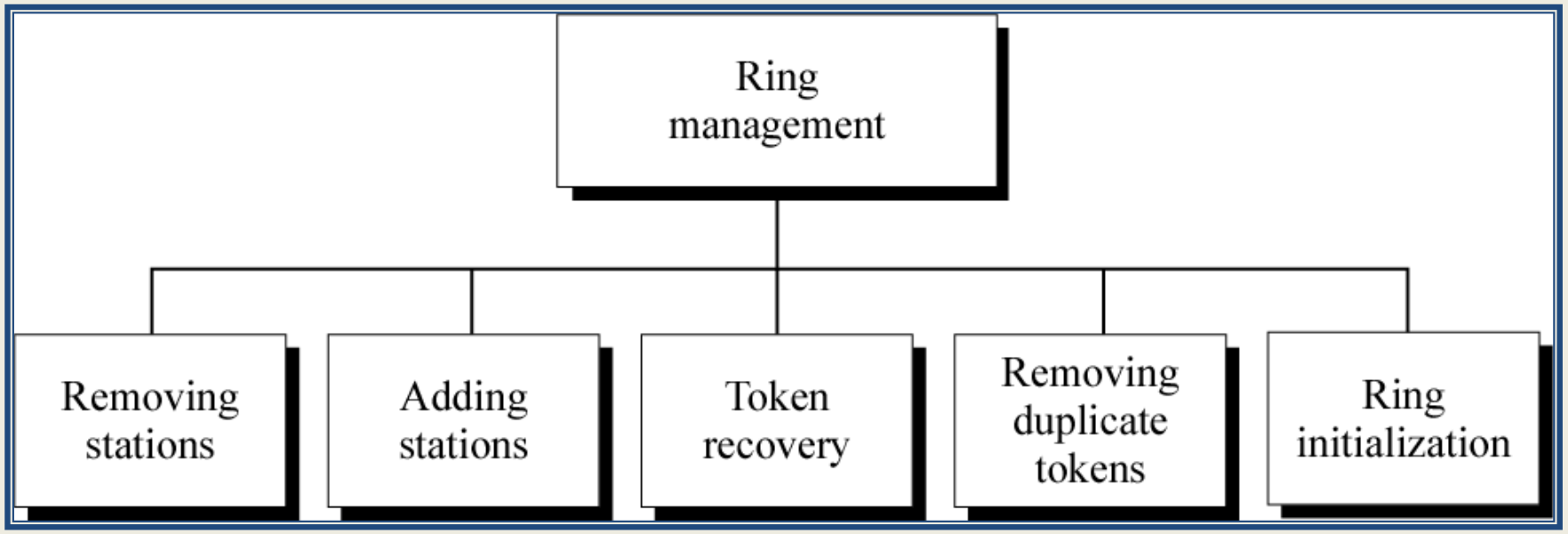
Contd...


112
70
20
112
70
90
45
c. Station 70 sends a data frame to station 112
112
70
20
45
90
45
d. Station 70 sends the token to station 45

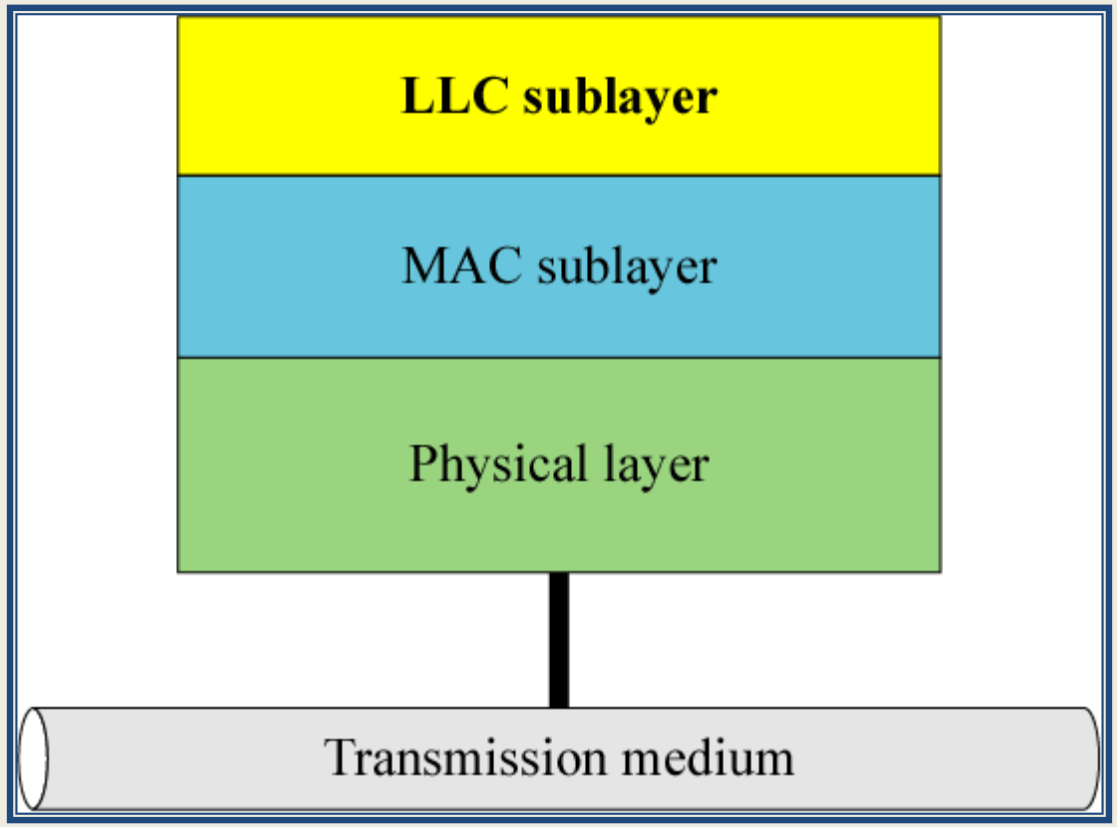
Queues for Service Classes



Ring Management



Token Bus Layers



Token Bus Frame

SD Start delimiter (flag)
FC Frame control (frame type)
ED End delimiter (flag)
FCS Frame Check Sequence

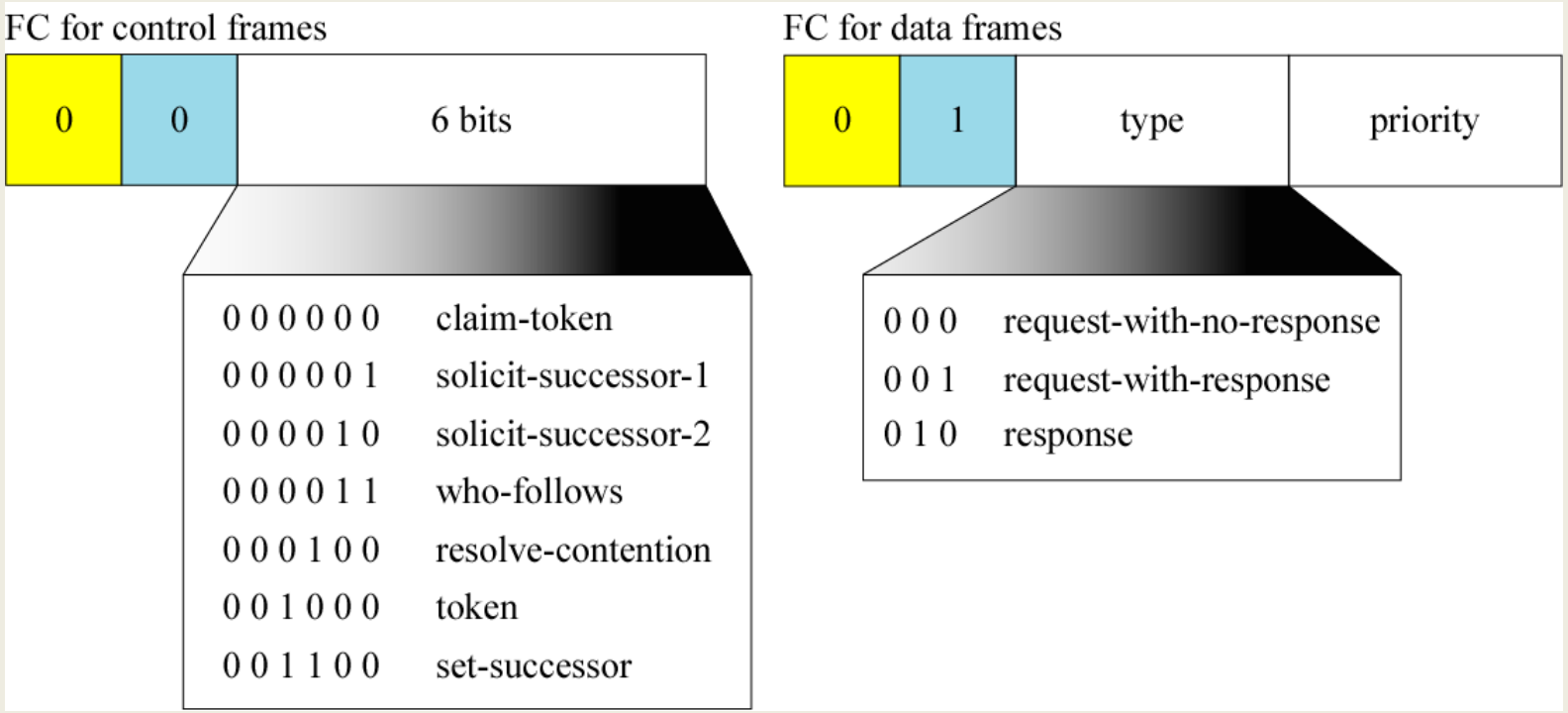
Preamble	SD	FC	Destination address	Source address	Data	FCS	ED
1 or more bytes	1 byte	1 byte	2 or 6 bytes	2 or 6 bytes	0 to 8190 bytes	4 bytes	1 byte

- Preamble: Establish bit synchronization and locate the first bit of the frame
- SD = Start Delimiter
- FC = Frame Control
- DA = Destination Address
- SA = Source Address
- FCS = Frame Check Sequence
- ED = End Delimiter
- Maximum Length (From SD to ED): 8191 Bytes

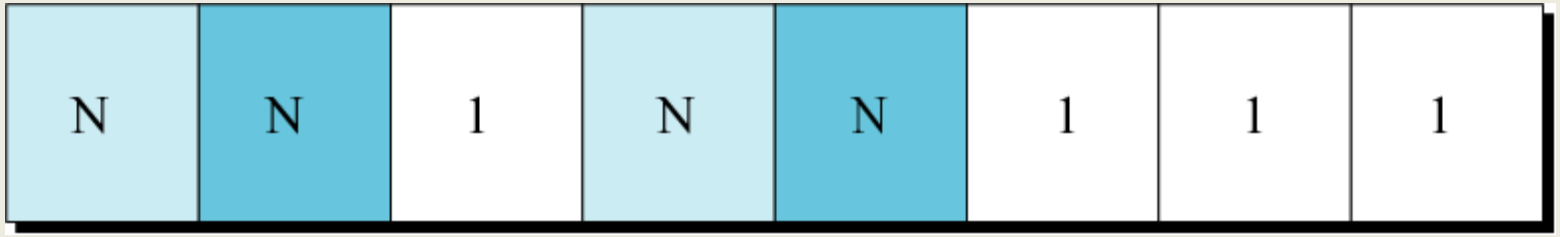
SD Field



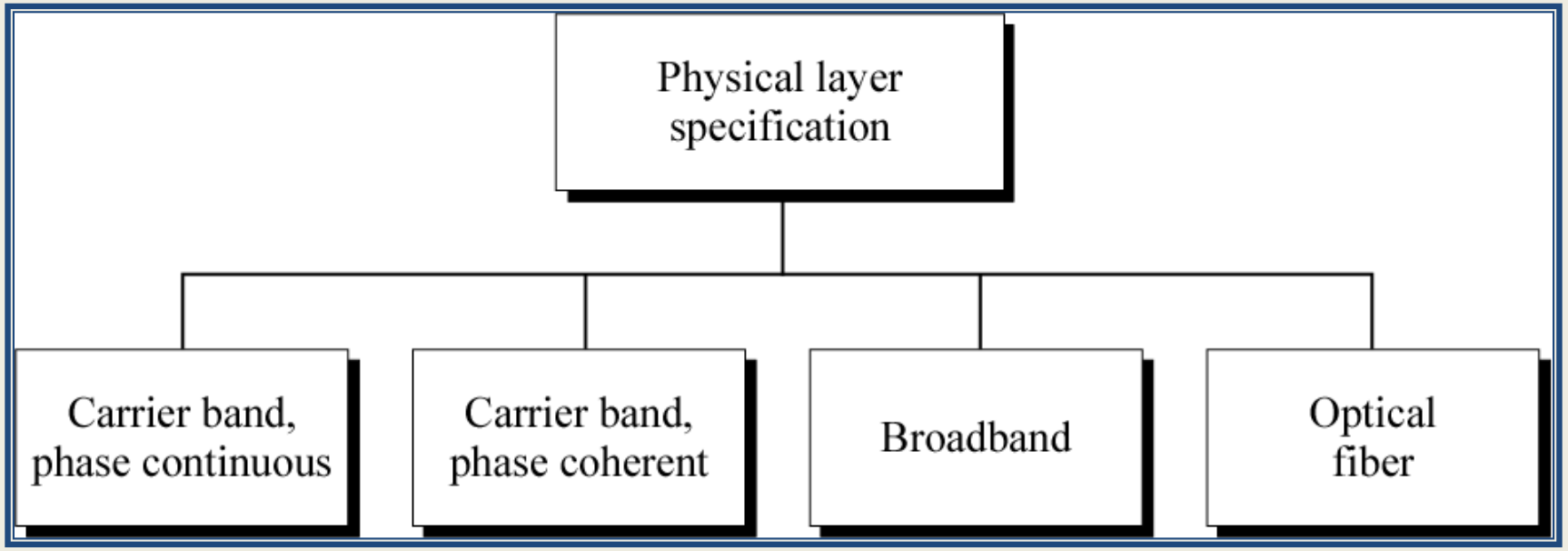
FC Field Format



ED Field Format



Physical Layer Specification



Token Ring

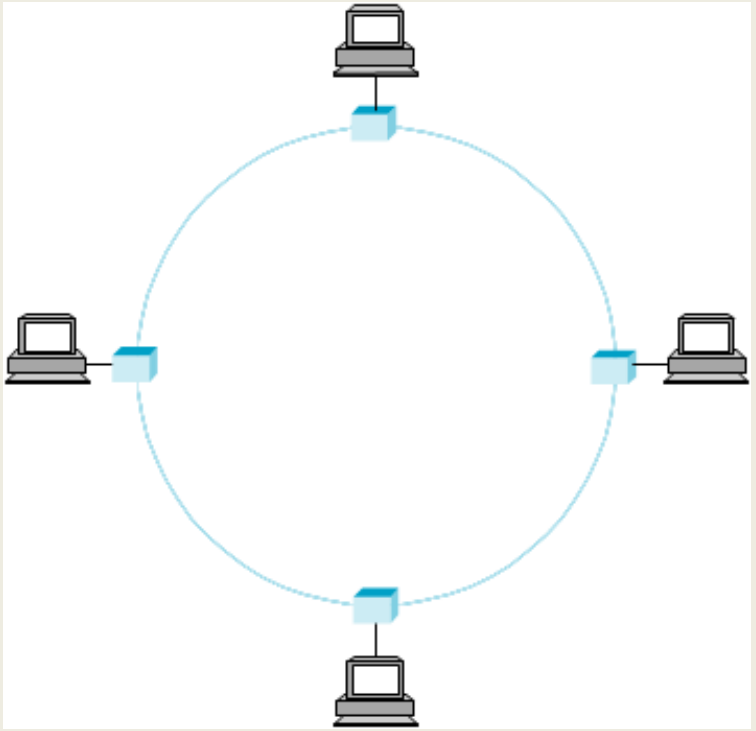
IEEE 802.5 standard

IEEE 802.5 (Token Ring): Ring Topology

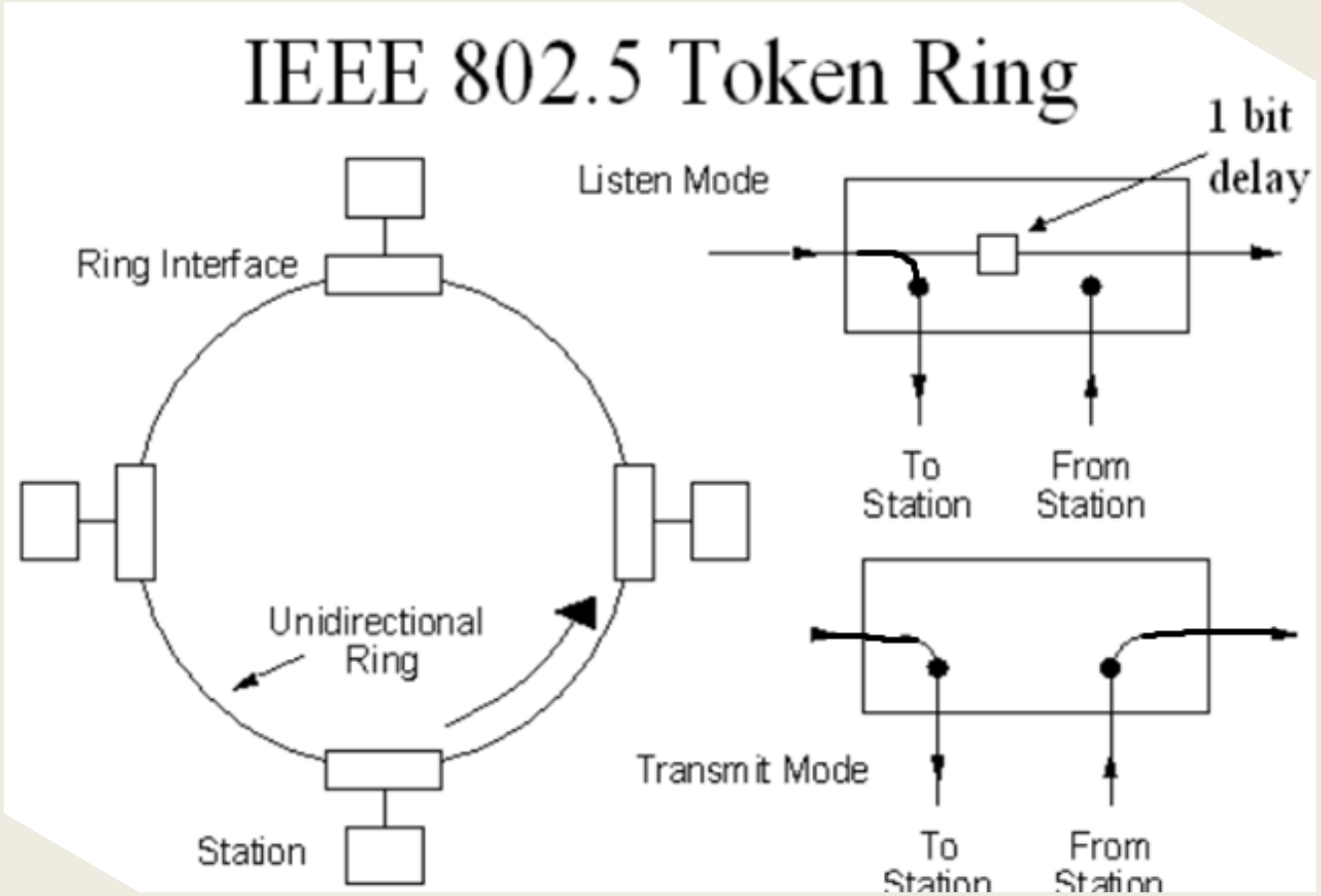
- Shared ring medium: all nodes see all frames
- Round Robin MAC Protocol: determines which station can transmit
- A special 3-byte pattern, the **token**, circulates around the ring perpetually and represents the "right to transmit"
- This establishes round-robin media access
- Data flow is unidirectional

Contd...

- All data flows in a particular direction around the ring; nodes receive frames from their upstream neighbor and forward them to their downstream neighbor
- Data rate: 4 or 16 Mbps



Contd...

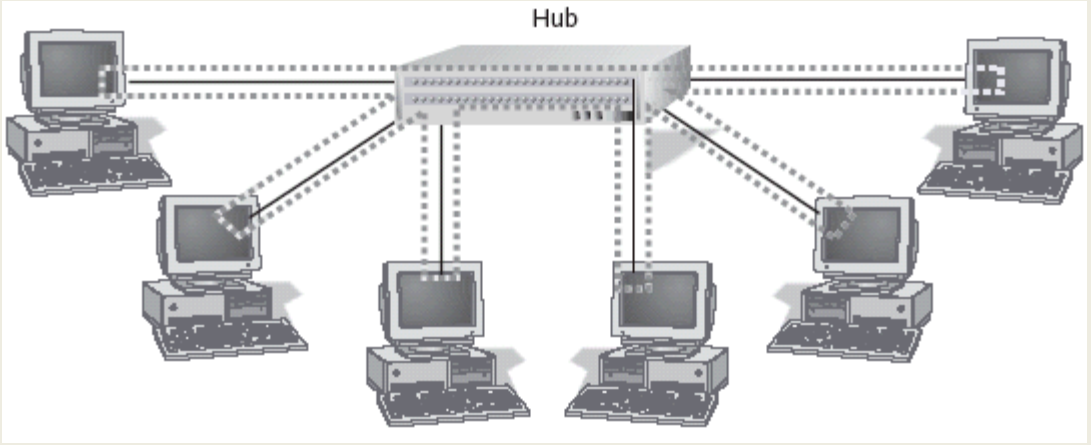

IEEE 802.5 Token Ring
Ring Interface
Unidirectional Ring
Station
Listen Mode
1 bit delay
To Station
From Station
Transmit Mode
To Station
From Station

IEEE 802.5 (Token Ring) Operation

- The token bit sequence circulates around the ring.
- Each station forwards the token if it does not have a frame to transmit.
- A station with data to send seizes the token (repeater now in transmit state) and begins sending it's frame. It can transmit for length of time called the **Token Hold Time (THT)** = 10 m seconds.
- Each station forwards the frame. The destination station notices its address and saves a copy of the frame as it also forwards the frame.
- When the sender sees its frame return, it drains it from the ring and reinserts a token. When the last bit of the returning frame has been drained, the repeater switches immediately to the listen state.

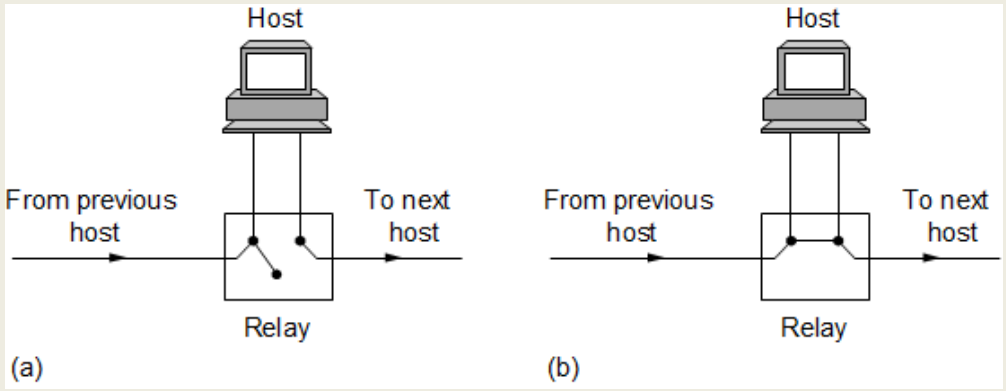
IEEE 802.5 (Token Ring) using a Hub

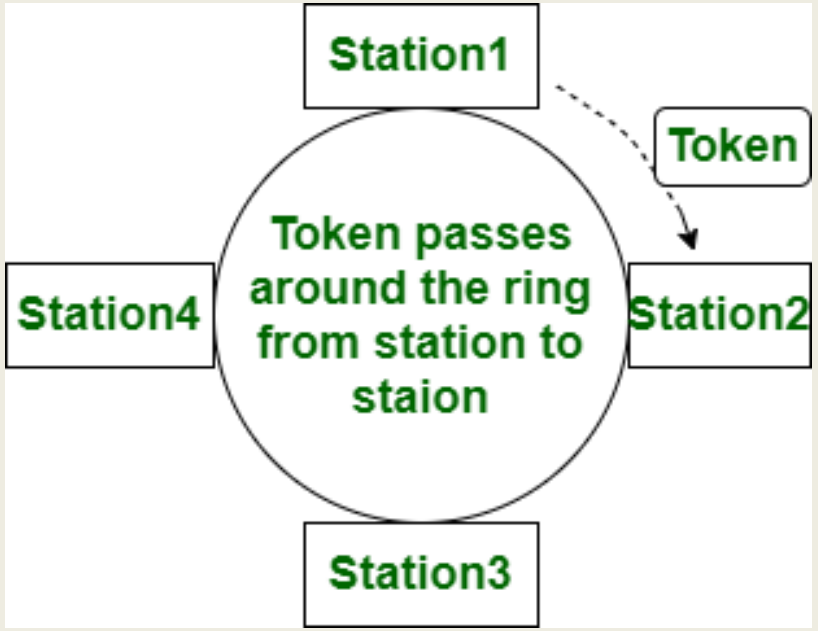
- The star-wired ring topology uses the physical layout of a star in conjunction with the token-passing data transmission method.
- Data are sent around the star in a circular pattern.
- This hybrid topology benefits from the fault tolerance of the star topology and the reliability of token passing.



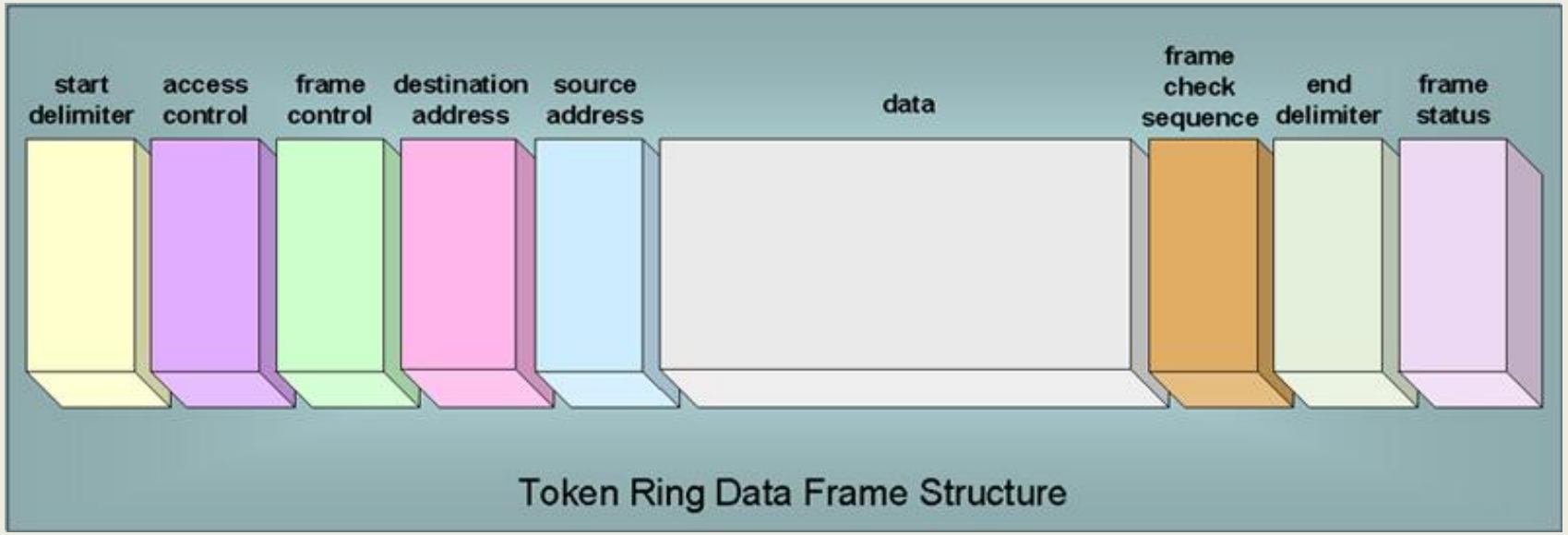
IEEE 802.5 (Token Ring): Bypass relays and Wire Center

- Bypass relays protect ring topology from node failure at the hardware level.





IEEE 802.5 (Token Ring) Frame Format



Start Delimiter	Access Control	End Delimiter

IEEE 802.5 (Token Ring) Frame Format

- Start delimiter (1 byte): serves the same basic purpose as the preamble in an Ethernet frame.
- Access Control (1 byte): contains the token bit, monitor bit, and priority bits
- Frame Control (1 byte): contains access control information
- Destination Address (6 bytes)
- Source Address (6 bytes)
- Data (no size limit specified): this is the actual data being sent (IP packet). Since the THT = 10 mseconds, practical

Contd...

- Size limit of the frame is 4500 bytes.
- Frame Check Sequence (4 bytes): CRC error checking bits
- End Delimiter (1 byte): signifies the end of the frame
- Frame Status Field (1 byte): serves as the ACK and indicates whether the address was recognized and the frame copied, which is done by the receiving computer before being sent back around the ring

IEEE 802.5 (Token Ring): Ring Maintenance

- There is a special station on the ring called a **monitor** station. It is responsible to identify and address situations dealing with a *lost token* and an *orphan frame*.
- Lost Token:
  - *Monitor station knows the number of stations on the ring and so calculates max*$THT = n * THT$ *(n is the number of stations on the ring). It keeps a timer of how long since it last saw the token pass by. If this is more than maxTRT, it drains the ring and inserts a new token in the ring.*

Contd...

- Orphan frame:
  - *A frame can get orphaned if the sending station goes down before it can drain it's frame.*
  - *As a frame passes by the monitor, it sets the "monitor" bit in the header of the frame.*
  - *If it sees a frame with this bit already set, it knows it is an orphan frame.*
  - *Then the monitor drains the ring and inserts a new token in the ring.*

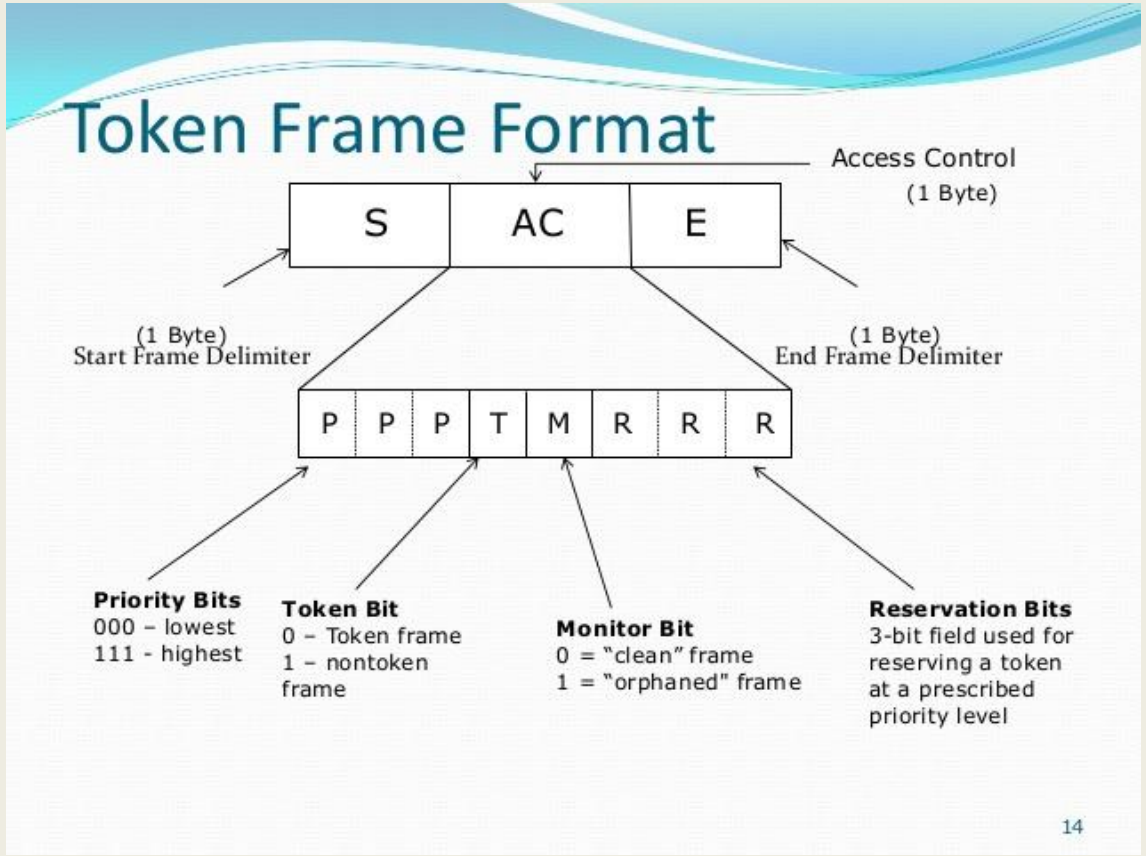
Ethernet vs Token Ring

- Ethernet is widely used at present (> 90% market share). People are experienced in using this technology.
- Ethernet uses CSMA/CD as the MAC protocol while Token Ring uses Round Robin protocol.
- Token Ring uses point-to-point connections between ring interfaces so that the electronic hardware can be fully digital and simple.
- There is no need for collision detection. The Ethernet NIC card requires some analog circuitry to be able to detect collisions.

Contd...

- Token Ring has excellent throughput at high loads since there is no possibility of collisions unlike 802.3.
- Under light load, Token Ring experiences token passing overhead. Ethernet has no such overhead and has excellent performance at light loads.

IEEE 802.5 (Token Ring) Token Frame Format



Frame Format Details

- The *access control* byte contains the priority and reservation fields, as well as a token bit (used to differentiate a token from a data/command frame) and a monitor bit (used by the active monitor to determine whether a frame is circling the ring endlessly).
- The *frame control* byte indicates whether the frame contains data or control information.
- In control frames, this byte specifies the type of control information.

Contd...

- The *frame status* byte is only present in Token Ring frames.
- It contains the A and C bits. When a frame arrives at the interface of a station with the destination address, the interface sets the A bit (=1), as it passes through.
- If the interface copies the frame to the station, it also sets the C bit (=1).
- A station might fail to copy a frame due to lack of buffer space or other reasons.

Contd...

- When the station which sent the frame strips it from the ring, it examines the A and C bits.The three possible combinations are;

1. A=0 and C=0; Destination not present or powered up.
2. A=1 and C=0; Destination present but frame not accepted.
3. A=1 and C=1; Destination present and frame copied.

- This arrangement provides an automatic acknowledgment of the delivery status of each frame.

IEEE 802.5 (Token Ring) Hierarchical Setup

